

Department of Information Engineering
MSc in Computer Engineering (a.a. 2022/2023)
Computer Architecture (Prof. Cosimo Antonio Prete)

Introduction To Quantum Computing

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Recap From Previous Seminar

Special 1-Qubit Gates

Gate	Action on Computational Basis	Matrix Representation
Identity	$I 0\rangle = 0\rangle$ $I 1\rangle = 1\rangle$	$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$
Pauli X	$X 0\rangle = 1\rangle$ $X 1\rangle = 0\rangle$	$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
Pauli Y	$Y 0\rangle = i 1\rangle$ $Y 1\rangle = -i 0\rangle$	$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$
Pauli Z	$Z 0\rangle = 0\rangle$ $Z 1\rangle = - 1\rangle$	$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$
Phase S	$S 0\rangle = 0\rangle$ $S 1\rangle = i 1\rangle$	$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$
T	$T 0\rangle = 0\rangle$ $T 1\rangle = e^{i\pi/4} 1\rangle$	$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$
Hadamard H	$H 0\rangle = \frac{1}{\sqrt{2}}(0\rangle + 1\rangle)$ $H 1\rangle = \frac{1}{\sqrt{2}}(0\rangle - 1\rangle)$	$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

Rotation About x-, y-, and z-Axes

$$R_x(\alpha) = \exp(-i\alpha X/2) = \begin{bmatrix} \cos\left(\frac{\alpha}{2}\right) & -i\sin\left(\frac{\alpha}{2}\right) \\ -i\sin\left(\frac{\alpha}{2}\right) & \cos\left(\frac{\alpha}{2}\right) \end{bmatrix}$$

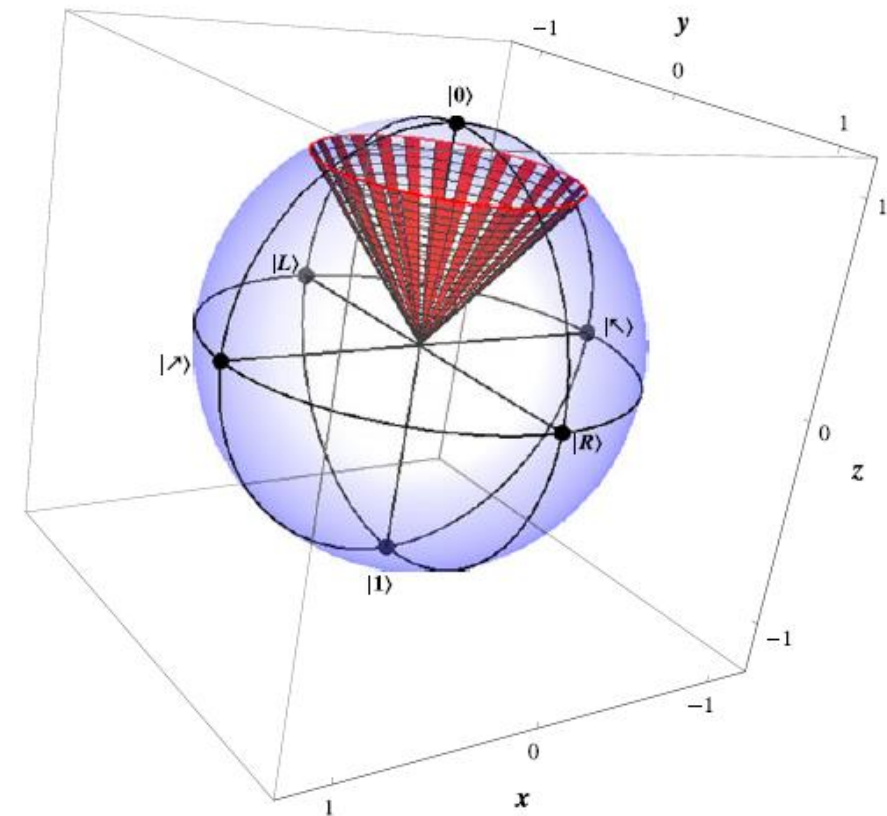
$$R_z(\alpha) = \exp(-i\alpha Z/2) = \begin{bmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{bmatrix}$$

$$R_y(\alpha) = \exp(-i\alpha Y/2) = \begin{bmatrix} \cos\left(\frac{\alpha}{2}\right) & -\sin\left(\frac{\alpha}{2}\right) \\ \sin\left(\frac{\alpha}{2}\right) & \cos\left(\frac{\alpha}{2}\right) \end{bmatrix}$$

$$Ph(\delta) = e^{i\delta} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

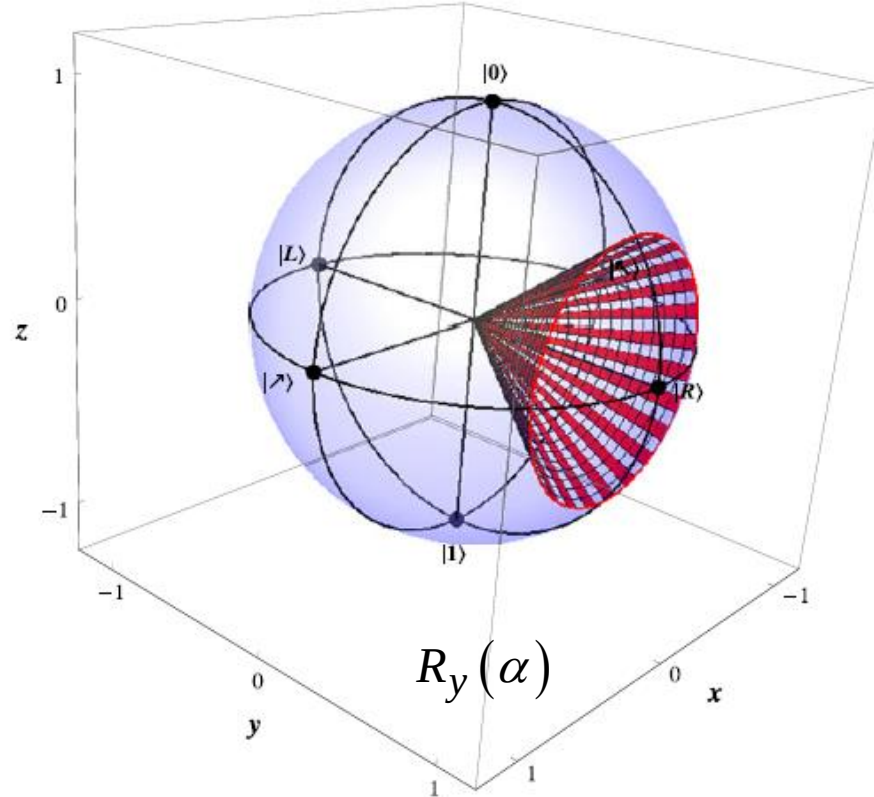
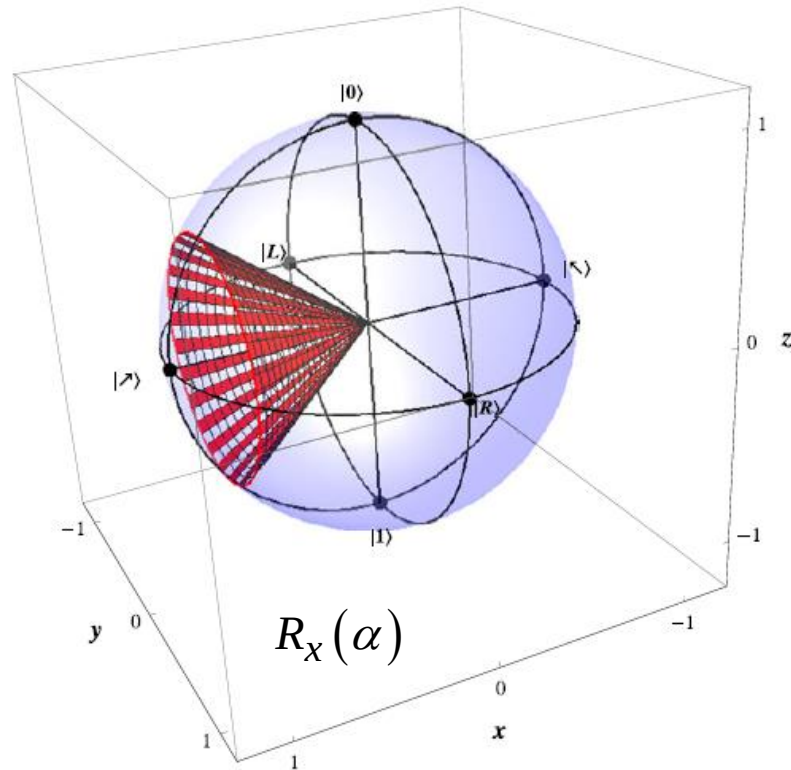
Rotation About x-, y-, and z-Axes

- The action of the $R_z(\alpha)$ gate on $|\psi\rangle$ has been to advance the angle ϕ by α and hence rotate the state about the z-axis through angle α
- This is why we call $R_z(\alpha)$ a z-rotation gate



Rotation About x-, y-, and z-Axes

- We leave it to the exercises for you to prove that $R_x(\alpha)$ and $R_y(\alpha)$ rotate the state about the x- and y-axes respectively



Rotation About x-, y-, and z-Axes

- Rotations on the Bloch sphere do not conform to commonsense intuitions about rotations that we have learned from our experience of the everyday world
- In particular, usually, a rotation of 2π radians (i.e., 360 degrees) of a solid object about any axis, restores that object to its initial orientation

Rotation About x-, y-, and z-Axes

- However, this is not true of rotations on the Bloch sphere! When we rotate a quantum state through 2π on the Bloch sphere we don't return it to its initial state
- Instead we pick up a phase factor

Rotation About x-, y-, and z-Axes

- To see this, let's compute the effect of rotating our arbitrary single qubit state

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle$$

about the z-axis through 2π radians

$$R_z(2\pi)|\psi\rangle = \begin{bmatrix} e^{-i\pi} & 0 \\ 0 & e^{i\pi} \end{bmatrix} \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) \\ e^{i\phi} \sin\left(\frac{\theta}{2}\right) \end{bmatrix} = \begin{bmatrix} -\cos\left(\frac{\theta}{2}\right) \\ -e^{i\phi} \sin\left(\frac{\theta}{2}\right) \end{bmatrix} = -|\psi\rangle$$

which has an extra overall phase of -1

Rotation About x-, y-, and z-Axes

- To restore a state back to its original form we need to rotate it through 4π on the Bloch sphere
- Have you ever encountered anything like this in your everyday world? You probably think not, but you'd be wrong!
- See the “Dirac’s Belt” or the “Belt Trick” video <https://www.youtube.com/watch?v=Vfh21o-JW9Q>

X From a Rotation Gate

- On the Bloch sphere, it can be shown that X is a rotation of 180° about the x-axis together with *global phase shifts* of 90°

$$R_x(\pi) \cdot Ph\left(\frac{\pi}{2}\right) = \begin{bmatrix} \cos\frac{\pi}{2} & -i\sin\frac{\pi}{2} \\ -i\sin\frac{\pi}{2} & \cos\frac{\pi}{2} \end{bmatrix} e^{i\pi/2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

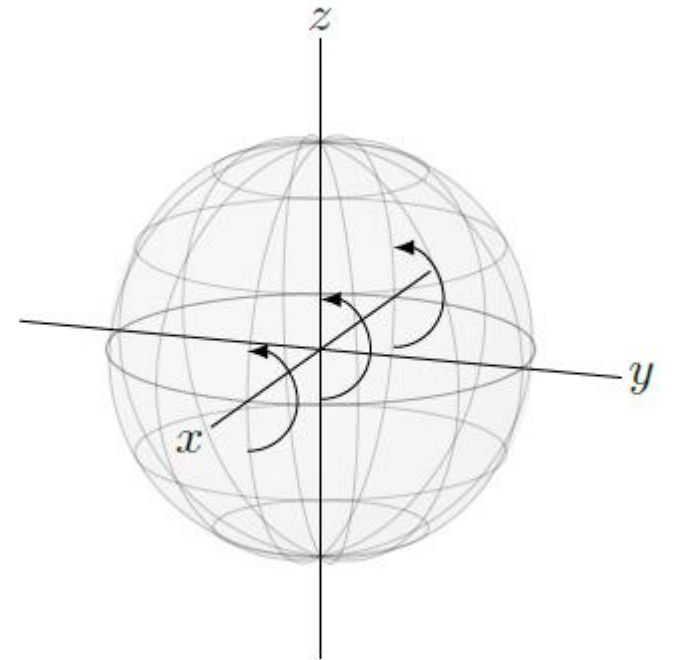
$$= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} (i) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = X$$

X From a Rotation Gate

- With this rotation in mind, we geometrically see that X causes $|0\rangle$ (the north pole) to rotate to $|1\rangle$ (the south pole), and vice versa
- We also see that $|i\rangle$ and $|-i\rangle$ rotate to each other, whereas $|+\rangle$ and $|-\rangle$ are unchanged
- Note, however, that mathematically

$$X|-\rangle = -|-\rangle \equiv |-\rangle$$

since the global phase does not matter

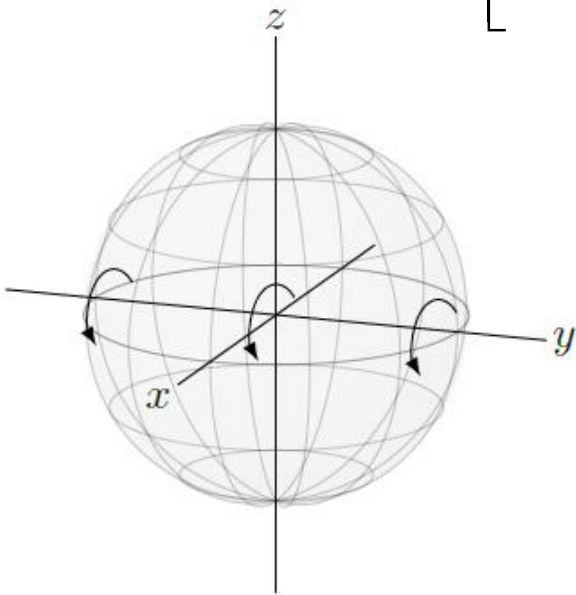


Y From a Rotation Gate

- On the Bloch sphere, it can be shown that Y is a rotation of 180° about the y -axis together with *global phase shifts* of 90°

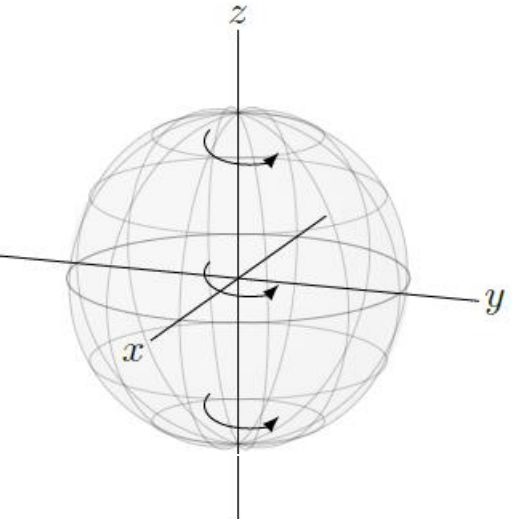
$$R_y(\pi) \cdot Ph\left(\frac{\pi}{2}\right) = \begin{bmatrix} \cos \frac{\pi}{2} & -\sin \frac{\pi}{2} \\ \sin \frac{\pi}{2} & \cos \frac{\pi}{2} \end{bmatrix} e^{i\pi/2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} (i) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = Y$$



Z From a Rotation Gate

- On the Bloch sphere, it can be shown that Z is a rotation of 180° about the z -axis together with *global phase shifts* of 90°

$$\begin{aligned}
 R_Z(\pi) \cdot Ph\left(\frac{\pi}{2}\right) &= \begin{bmatrix} e^{-i\pi/2} & 0 \\ 0 & e^{i\pi/2} \end{bmatrix} e^{i\pi/2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\frac{\pi}{2} - i\sin\frac{\pi}{2} & 0 \\ 0 & \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \end{bmatrix} \left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} \left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} (i) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = Z
 \end{aligned}$$


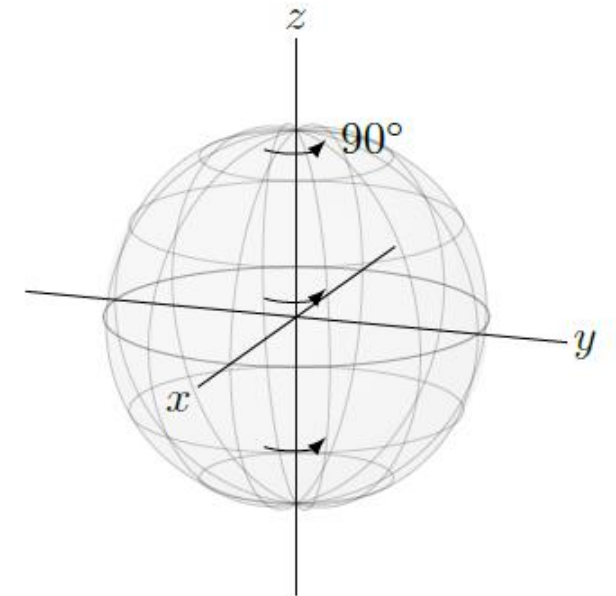
S From a Rotation Gate

- On the Bloch sphere, it can be shown that S is a rotation of 90° about the z -axis together with *global phase shifts* of 45°

$$R_z\left(\frac{\pi}{2}\right) \cdot Ph\left(\frac{\pi}{4}\right) = \begin{bmatrix} e^{-i\pi/4} & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} e^{i\pi/8} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \cos\frac{\pi}{4} - i\sin\frac{\pi}{4} & 0 \\ 0 & \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} \end{bmatrix} \left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \begin{bmatrix} 1-i & 0 \\ 0 & 1+i \end{bmatrix} \left(\frac{1}{\sqrt{2}} (1+i) \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

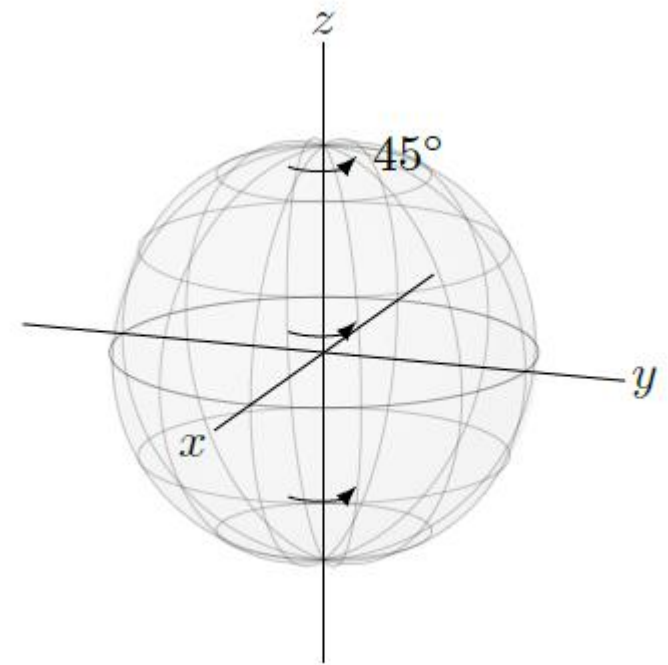
$$= \frac{1}{2} \begin{bmatrix} 1-i & 0 \\ 0 & 1+i \end{bmatrix} \begin{bmatrix} 1+i & 0 \\ 0 & 1+i \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1-i^2 & 0 \\ 0 & (1+i)^2 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 2i \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} = S$$



T From a Rotation Gate

- On the Bloch sphere, it can be shown that T (also called $\pi/8$ gate) is a rotation of 45° about the z-axis together with *global phase shifts* of $\pi/8$ radians

$$\begin{aligned} R_z\left(\frac{\pi}{4}\right) \cdot Ph\left(\frac{\pi}{8}\right) &= \begin{bmatrix} e^{-i\pi/8} & 0 \\ 0 & e^{i\pi/8} \end{bmatrix} e^{i\pi/8} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} = T \end{aligned}$$



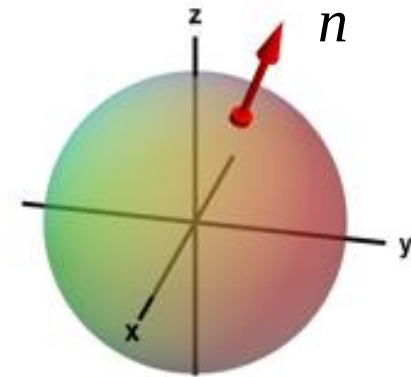
H From Rotation Gates

$$\begin{aligned} R_x(\pi) \cdot R_y\left(\frac{\pi}{2}\right) \cdot Ph\left(\frac{\pi}{2}\right) &= \begin{bmatrix} \cos \frac{\pi}{2} & -i \sin \frac{\pi}{2} \\ -i \sin \frac{\pi}{2} & \cos \frac{\pi}{2} \end{bmatrix} \begin{bmatrix} \cos \frac{\pi}{4} & -\sin \frac{\pi}{4} \\ \sin \frac{\pi}{4} & \cos \frac{\pi}{4} \end{bmatrix} e^{i\pi/2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2} \right) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{i}{\sqrt{2}} \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = H \end{aligned}$$

Rotation About an Arbitrary Axis

- If $n = [n_x, n_y, n_z]$ is a real unit vector in three dimensions, then it can be shown that the operator $R_n(\theta)$ rotates the Bloch vector by an angle θ about the n axis, where

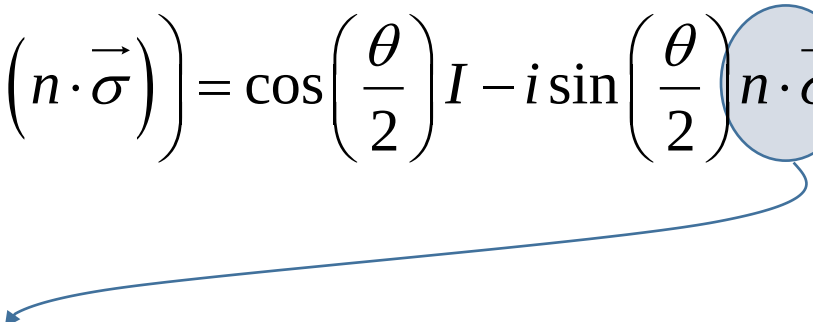
$$R_n(\theta) = \exp\left(-i\theta\left(n \cdot \frac{\vec{\sigma}}{2}\right)\right)$$



and $\vec{\sigma}$ denotes the three component vector (X,Y,Z) of Pauli matrices

Rotation About an Arbitrary Axis

- Furthermore, it is not hard to show that $(\vec{n} \cdot \vec{\sigma})^2 = I$, and therefore we can use the special case operator exponential and write

$$R_n(\theta) = \exp\left(-i\theta\left(\vec{n} \cdot \frac{\vec{\sigma}}{2}\right)\right) = \exp\left(-i\frac{\theta}{2}(\vec{n} \cdot \vec{\sigma})\right) = \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)(\vec{n} \cdot \vec{\sigma})$$
$$= \cos\left(\frac{\theta}{2}\right)I - i\sin\left(\frac{\theta}{2}\right)(n_x X + n_y Y + n_z Z)$$


Given a real number x and an A matrix such that $A^2 = -I$ the following relation holds



$$e^{iAx} = \cos(x)I + i\sin(x)A$$

Arbitrary Unitary Operator

- It can be shown that *an arbitrary single qubit unitary operator* U can be written in the form

$$U = \exp(i\alpha) R_n(\theta)$$

for some real number α and θ , and a real three-dimensional unit vector n



$$U = \exp(i\alpha) R_n(\theta) = \exp(i\alpha) \left[\cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) (n_x X + n_y Y + n_z Z) \right]$$

Arbitrary Unitary Operator

$$U = \exp(i\alpha) \left[\cos\left(\frac{\theta}{2}\right) I - i \sin\left(\frac{\theta}{2}\right) (n_x X + n_y Y + n_z Z) \right]$$

- For example, consider

$$\alpha = \frac{\pi}{2}, \quad \theta = \pi, \quad \text{and} \quad n = \left[\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right] \quad \longrightarrow \quad n \cdot \vec{\sigma} = (n_x X + n_y Y + n_z Z) = \frac{X + Z}{\sqrt{2}}$$

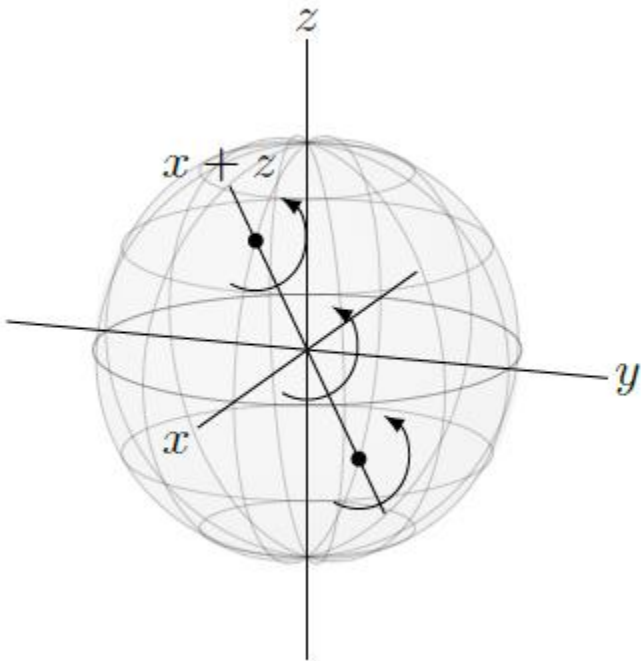
$$U = \exp(i\pi/2) R_n(\pi) = \exp(\cancel{i\pi/2}) \left[\cos\left(\frac{\pi}{2}\right) I - i \sin\left(\frac{\pi}{2}\right) \frac{1}{\sqrt{2}} (X + Z) \right]$$

$$= \left[\cos\left(\frac{\pi}{2}\right) I - i \sin\left(\frac{\pi}{2}\right) \frac{1}{\sqrt{2}} (X + Z) \right] = \cancel{-} \frac{1}{\sqrt{2}} (X + Z)$$

$$= \frac{1}{\sqrt{2}} (X + Z) = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = H$$

H From a Rotation Gate

Thus, on the Bloch sphere, H is a rotation of 180° about the $x+z$ -axis together with *global phase shifts* of 90°



Arbitrary Unitary Operator

- An arbitrary unitary operator on a single qubit can be written in many ways as a combination of rotations, together with *global phase shifts* on the qubit
- For example, we have already seen that

$$X = R_x(\pi) \cdot Ph\left(\frac{\pi}{2}\right) \qquad H = R_x(\pi) \cdot R_y\left(\frac{\pi}{2}\right) \cdot Ph\left(\frac{\pi}{2}\right)$$

- However, we can easily show that

$$X = R_y(\pi) \cdot R_z(\pi) \cdot Ph\left(\frac{\pi}{2}\right) \qquad H = R_y\left(\frac{\pi}{2}\right) \cdot R_z(\pi) \cdot Ph\left(\frac{\pi}{2}\right)$$

Arbitrary Unitary Operator

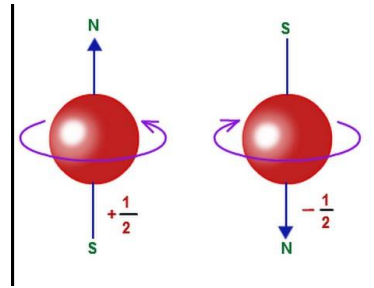
- The following theorem will be particularly useful in later applications to controlled operations.
- *Theorem: (Z-Y **decomposition for a single qubit**)* Suppose U is a unitary operation on a single qubit. Then there exist real numbers α , β , γ and δ such that

$$U = \exp(i\alpha) R_z(\beta) \cdot R_y(\gamma) \cdot R_z(\delta)$$

Composite Systems

Composite Systems

- Let's consider a composite quantum system made up of two distinct qubits, also referred to as a *bipartite* system

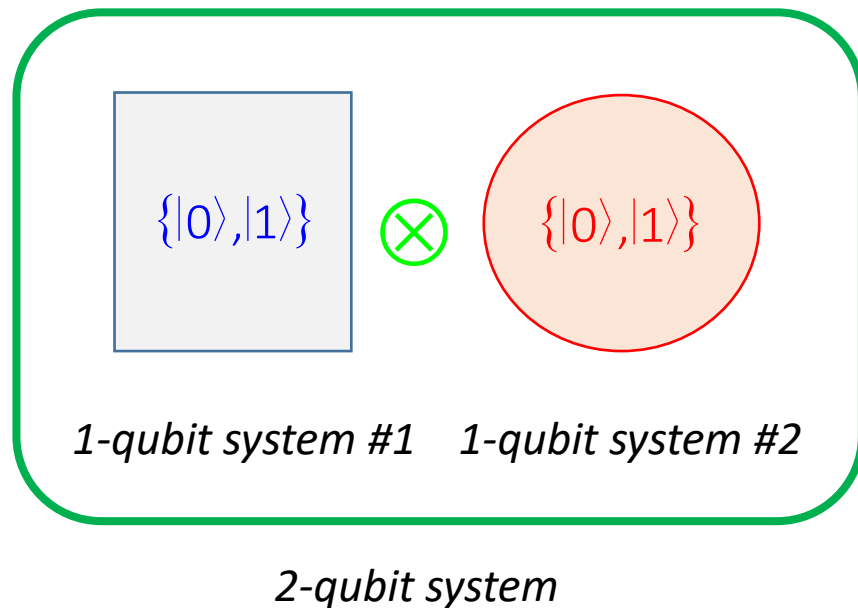


- How should we describe states of the composite system?
- **Postulate 4** describes how the state space of a composite system is built up from the state spaces of the component systems

Composite Systems

Postulate 4:

- The state space of a *composite physical system* is the *tensor product* of the state spaces of the *component physical systems*.



Moreover, if we have systems numbered 1 through n , and system number i is prepared in the state $|\psi_i\rangle$, then the joint state of the total system is $|\psi_1\rangle \otimes |\psi_2\rangle \otimes \cdots \otimes |\psi_n\rangle$



Thus, the tensor product is the mathematical structure used to describe the state space of a composite physical system

Tensor Products

- The *tensor product* is a way of putting vector spaces together to form larger vector spaces
- Suppose V and W are Hilbert spaces of dimension m and n respectively
- Then $V \otimes W$ (read “ V tensor W ”) is an mn dimensional vector space
- Therefore the dimension of the Hilbert space describing the states of the system composed of 2 qubits is $2 \times 2 = 2^2 = 4$
- If the number of qubits is k , the dimension is 2^k

Tensor Products

- The elements of $V \otimes W$ are linear combinations of *tensor products* $|v\rangle \otimes |w\rangle$ of elements $|v\rangle \in V$ and $|w\rangle \in W$
- We often use the abbreviated notations $|v\rangle|w\rangle$ or even $|vw\rangle$ for the tensor product $|v\rangle \otimes |w\rangle$
- If $\{|0\rangle, |1\rangle\}$ is the standard basis for V , then the standard basis for $V \otimes V$ is

$$\{|0\rangle \otimes |0\rangle, |0\rangle \otimes |1\rangle, |1\rangle \otimes |0\rangle, |1\rangle \otimes |1\rangle\}$$

$$\{|0\rangle|0\rangle, |0\rangle|1\rangle, |1\rangle|0\rangle, |1\rangle|1\rangle\}$$

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$$

Equivalent notations

Tensor Products

- For example, if V is a two-dimensional vector space with basis vectors $|0\rangle$ and $|1\rangle$ then

$$\frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |0\rangle)$$

is an element of $V \otimes V$

Tensor Products

- In linear algebra, the *tensor product* is simply the *Kronecker product*, which is obtained by multiplying each term of the *first* matrix/vector by the *entire second* matrix/vector
- Specifically, with two qubits $|\psi\rangle = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$ and $|\varphi\rangle = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}$

$$|\psi\rangle \otimes |\varphi\rangle = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} \otimes \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} = \begin{bmatrix} a_0 \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} \\ a_1 \begin{bmatrix} b_0 \\ b_1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} a_0 b_0 \\ a_0 b_1 \\ a_1 b_0 \\ a_1 b_1 \end{bmatrix} \xrightarrow{\text{Example}} |0\rangle \otimes |1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ 0 \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

Tensor Products

- An additional example of tensor product between two matrices

$$X \otimes Y = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} 0 \times \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} & 1 \times \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \\ 1 \times \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} & 0 \times \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{bmatrix}$$

Useful Properties

$$(A \otimes B)^* = A^* \otimes B^* \quad \text{Property \#1}$$

$$(A \otimes B)^T = A^T \otimes B^T \quad \text{Property \#2}$$

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger \quad \text{Property \#3}$$

$$(A \otimes B) \cdot (C \otimes D) = (AC) \otimes (BD) \quad \text{Property \#4}$$

Proofs of the above-mentioned equalities will be shown later under the assumption that A , B , C and D are $n \times n$ matrices

Tensor Products Properties

By definition the tensor product satisfies the following basic properties:

1. For an arbitrary scalar z (*complex number*) and elements $|v\rangle$ of V and $|w\rangle$ of W

$$z(|v\rangle \otimes |w\rangle) = (z|v\rangle) \otimes |w\rangle = |v\rangle \otimes (z|w\rangle)$$

1. For arbitrary $|v_1\rangle$ and $|v_2\rangle$ in V and $|w\rangle$ in W ,

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle$$

2. For arbitrary $|v\rangle$ in V and $|w_1\rangle$ and $|w_2\rangle$ in W ,

$$|v\rangle \otimes (|w_1\rangle + |w_2\rangle) = |v\rangle \otimes |w_1\rangle + |v\rangle \otimes |w_2\rangle$$

Tensor Products Properties (Example)

By exploiting the above properties, if:

$$|v\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \quad |w\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$(|v_1\rangle + |v_2\rangle) \otimes |w\rangle = |v_1\rangle \otimes |w\rangle + |v_2\rangle \otimes |w\rangle$$

Then

$$|v\rangle \otimes |w\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{2} [|0\rangle \otimes (|0\rangle - |1\rangle) + |1\rangle \otimes (|0\rangle - |1\rangle)]$$

$$= \frac{1}{2} (|0\rangle \otimes |0\rangle - |0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle - |1\rangle \otimes |1\rangle)$$

$$|v\rangle \otimes (|w_1\rangle + |w_2\rangle) = |v\rangle \otimes |w_1\rangle + |v\rangle \otimes |w_2\rangle$$

Orthonormal Basis

- If

$$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$$

is the standard basis for V , then we are going to derive the vector form of the standard basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ for the composite system $V \otimes V$

$$|00\rangle = |0\rangle|0\rangle = |0\rangle \otimes |0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|01\rangle = |0\rangle|1\rangle = |0\rangle \otimes |1\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$|10\rangle = |1\rangle|0\rangle = |1\rangle \otimes |0\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ 1 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$|11\rangle = |1\rangle|1\rangle = |1\rangle \otimes |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ 1 \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

Orthonormal Basis for $V \otimes V$

- It's easy to show that $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ *is a complete orthonormal basis* for the state of two qubits by showing that it satisfies the completeness relation

$$|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + |11\rangle\langle 11| = I$$

$$|00\rangle\langle 00| = (|0\rangle \otimes |0\rangle)(\langle 0| \otimes \langle 0|) = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)^\dagger = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Orthonormal Basis for $V \otimes V$

$$|01\rangle\langle 01| = (|0\rangle \otimes |1\rangle)(\langle 0| \otimes \langle 1|) = \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)^\dagger = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$|10\rangle\langle 10| = (|1\rangle \otimes |0\rangle)(\langle 1| \otimes \langle 0|) = \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right) \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)^\dagger = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$|11\rangle\langle 11| = (|1\rangle \otimes |1\rangle)(\langle 1| \otimes \langle 1|) = \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right) \left(\begin{bmatrix} 0 \\ 1 \end{bmatrix} \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)^\dagger = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Orthonormal Basis for $V \otimes V$

- By summing up the previous results we obtain

$$|00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + |11\rangle\langle 11| = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I$$

Superposition

- As any other *quantum system*, a system composed by a pair of qubits can exist in *superposition* of the four states belonging to the standard basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ **(Postulate #1)**

$$|\psi\rangle = c_0 \begin{array}{c} \uparrow \\ \text{0} \\ \downarrow \end{array} \begin{array}{c} \uparrow \\ \text{0} \\ \downarrow \end{array} + c_1 \begin{array}{c} \uparrow \\ \text{0} \\ \downarrow \end{array} \begin{array}{c} \downarrow \\ \text{1} \\ \downarrow \end{array} + c_2 \begin{array}{c} \downarrow \\ \text{1} \\ \downarrow \end{array} \begin{array}{c} \uparrow \\ \text{0} \\ \downarrow \end{array} + c_3 \begin{array}{c} \downarrow \\ \text{1} \\ \downarrow \end{array} \begin{array}{c} \downarrow \\ \text{1} \\ \downarrow \end{array}$$

$$|\psi\rangle = c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle \quad \text{where} \quad |c_0|^2 + |c_1|^2 + |c_2|^2 + |c_3|^2 = 1$$

$$\frac{1}{\sqrt{2}}(|0\rangle \otimes |1\rangle + |1\rangle \otimes |0\rangle) \quad \text{Specific state with} \quad \begin{cases} c_0 = c_3 = 0 \\ c_1 = c_2 = 1/\sqrt{2} \end{cases}$$

Equivalent Notations

- We often use the abbreviated notations $|v\rangle|w\rangle$, $|v, w\rangle$ or even $|vw\rangle$ for the tensor product $|v\rangle \otimes |w\rangle$
- The notation admits the possibility of labeling each of the component kets with the vector space from whence it came, V or W . For example

$$\{|0_v 0_w\rangle, |0_v 1_w\rangle, |1_v 0_w\rangle, |1_v 1_w\rangle\}$$

is equivalent to

$$\{|0_v\rangle \otimes |0_w\rangle, |0_v\rangle \otimes |1_w\rangle, |1_v\rangle \otimes |0_w\rangle, |1_v\rangle \otimes |1_w\rangle\}$$

Equivalent Notations

- Alternative shorthands

$$|0\rangle \otimes |0\rangle \longleftrightarrow |0\rangle |0\rangle \longleftrightarrow |00\rangle \longleftrightarrow |0\rangle^2$$

$$|0\rangle \otimes |1\rangle \longleftrightarrow |0\rangle |1\rangle \longleftrightarrow |01\rangle \longleftrightarrow |1\rangle^2$$

$$|1\rangle \otimes |0\rangle \longleftrightarrow |1\rangle |0\rangle \longleftrightarrow |10\rangle \longleftrightarrow |2\rangle^2$$

$$|1\rangle \otimes |1\rangle \longleftrightarrow |1\rangle |1\rangle \longleftrightarrow |11\rangle \longleftrightarrow |3\rangle^2$$

Tensor Product for Bra

- The tensor product also works for *bras*, so

$$\langle v| \otimes \langle w| = \langle v| \langle w| = \langle vw|$$

- Furthermore,

$$|vw\rangle^\dagger = (|v\rangle \otimes |w\rangle)^\dagger = |v\rangle^\dagger \otimes |w\rangle^\dagger = \langle v| \otimes \langle w| = \langle v| \langle w| = \langle vw|$$

$$\longrightarrow |vw\rangle^\dagger = \langle vw|$$

Tensor Product for Bra

$$(A \otimes B) \cdot (C \otimes D) = (AC) \otimes (BD)$$

- Then, the inner product of, say $\langle vw|$ and $|rs\rangle$ is defined by

$$\langle vw|rs\rangle = (\langle v| \otimes \langle w|)(|r\rangle \otimes |s\rangle) = \langle v|r\rangle \langle w|s\rangle, \quad \forall |v\rangle, |r\rangle \in V \text{ and } \forall |w\rangle, |s\rangle \in W$$

- For qubit states $|00\rangle$ and $|01\rangle$

$$\langle \textcolor{red}{0}1|\textcolor{red}{0}0\rangle = \underbrace{\langle \textcolor{red}{0}| \textcolor{red}{0}\rangle}_1 \cdot \underbrace{\langle 1|0\rangle}_0 = 0 \quad \longrightarrow \quad \text{So, } |00\rangle \text{ and } |01\rangle \text{ are orthogonal}$$

Useful Properties

$$(A \otimes B)^* = A^* \otimes B^* \quad \text{Property \#1}$$

$$(A \otimes B)^T = A^T \otimes B^T \quad \text{Property \#2}$$

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger \quad \text{Property \#3}$$

$$(A \otimes B) \cdot (C \otimes D) = (AC) \otimes (BD) \quad \text{Property \#4}$$

It follows the proofs under the assumption that A , B , C and D are $n \times n$ matrices

Useful Properties

$$(A \otimes B)^* = A^* \otimes B^*$$

*definition of tensor product
between two matrices*

definition of conjugate matrix

Proof

$$\begin{aligned}
 (A \otimes B)^* &= \left(\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{bmatrix} \otimes B \right)^* \\
 &= \begin{bmatrix} A_{11}B & A_{12}B & \cdots & A_{1n}B \\ A_{21}B & A_{22}B & \cdots & A_{2n}B \\ \cdots & \cdots & \cdots & \cdots \\ A_{n1}B & A_{n2}B & \cdots & A_{nn}B \end{bmatrix}^* \\
 &= \begin{bmatrix} A_{11}^*B^* & A_{12}^*B^* & \cdots & A_{1n}^*B^* \\ A_{21}^*B^* & A_{22}^*B^* & \cdots & A_{2n}^*B^* \\ \cdots & \cdots & \cdots & \cdots \\ A_{n1}^*B^* & A_{n2}^*B^* & \cdots & A_{nn}^*B^* \end{bmatrix} \\
 &= \begin{bmatrix} A_{11}^* & A_{12}^* & \cdots & A_{1n}^* \\ A_{21}^* & A_{22}^* & \cdots & A_{2n}^* \\ \cdots & \cdots & \cdots & \cdots \\ A_{n1}^* & A_{n2}^* & \cdots & A_{nn}^* \end{bmatrix} \otimes B^* \\
 &= \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{bmatrix}^* \otimes B^* = A^* \otimes B^*
 \end{aligned}$$

Useful Properties

$$(A \otimes B)^T = A^T \otimes B^T$$

*definition of tensor product
between two matrices*

definition of transpose matrix

Proof

$$\begin{aligned}
 (A \otimes B)^T &= \left(\begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1n} \\ A_{21} & A_{22} & \cdots & A_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ A_{n1} & A_{n2} & \cdots & A_{nn} \end{bmatrix} \otimes B \right)^T = \left(\begin{bmatrix} A_{11}B & A_{12}B & \cdots & A_{1n}B \\ A_{21}B & A_{22}B & \cdots & A_{2n}B \\ \cdots & \cdots & \cdots & \cdots \\ A_{n1}B & A_{n2}B & \cdots & A_{nn}B \end{bmatrix} \right)^T = \begin{bmatrix} A_{11}B^T & A_{21}B^T & \cdots & A_{n1}B^T \\ A_{12}B^T & A_{22}B^T & \cdots & A_{n2}B^T \\ \cdots & \cdots & \cdots & \cdots \\ A_{1n}B^T & A_{2n}B^T & \cdots & A_{nn}B^T \end{bmatrix} \\
 &= \begin{bmatrix} A_{11}B^T & A_{21}B^T & \cdots & A_{n1}B^T \\ A_{12}B^T & A_{22}B^T & \cdots & A_{n2}B^T \\ \cdots & \cdots & \cdots & \cdots \\ A_{1n}B^T & A_{2n}B^T & \cdots & A_{nn}B^T \end{bmatrix} = \begin{bmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \cdots & \cdots & \cdots & \cdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{bmatrix} \otimes B^T = A^T \otimes B^T
 \end{aligned}$$

Useful Properties

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$$

Proof

$$(A \otimes B)^\dagger = \left((A \otimes B)^* \right)^T \stackrel{\text{Property \#1}}{=} \left(A^* \otimes B^* \right)^T \stackrel{\text{Property \#2}}{=} \left(A^* \right)^T \otimes \left(B^* \right)^T \stackrel{\text{definition of conjugate transpose (dagger)}}{=} A^\dagger \otimes B^\dagger$$

definition of conjugate transpose (dagger)

definition of conjugate transpose (dagger)

Useful Properties

$$(A \otimes B) \cdot (C \otimes D) = (AC) \otimes (BD)$$

Proof

- We will use the notation $(A \otimes B)_{ij} = (a_{ij} B)$ to denote block matrices.
Then

$$\begin{aligned} ((A \otimes B) \cdot (C \otimes D))_{ij} &= \sum_k (A \otimes B)_{ik} (C \otimes D)_{kj} = \sum_k (a_{ik} B) (c_{kj} D) \\ &= \sum_k (a_{ik} c_{kj}) (BD) = (AC)_{ij} (BD) = (AC \otimes BD)_{ij} \end{aligned}$$

Linear Operators on the Space $V \otimes W$

- Suppose $|v\rangle$ and $|w\rangle$ are vectors in V and W , and A and B are linear operators on V and W , respectively, i.e.

$$A:V \rightarrow V \qquad B:W \rightarrow W$$

- Then we can define a linear operator

$$A \otimes B:V \otimes W \rightarrow V \otimes W$$

by the equation

$$(A \otimes B)(|v\rangle \otimes |w\rangle) \equiv A|v\rangle \otimes B|w\rangle$$

Notations

- Finally, we mention the useful notation

$$|\psi\rangle^{\otimes k} = \underbrace{|\psi\rangle \otimes |\psi\rangle \otimes \dots \otimes |\psi\rangle}_{k \text{ times}}$$

- For example

$$|\psi\rangle^{\otimes 2} = |\psi\rangle \otimes |\psi\rangle$$

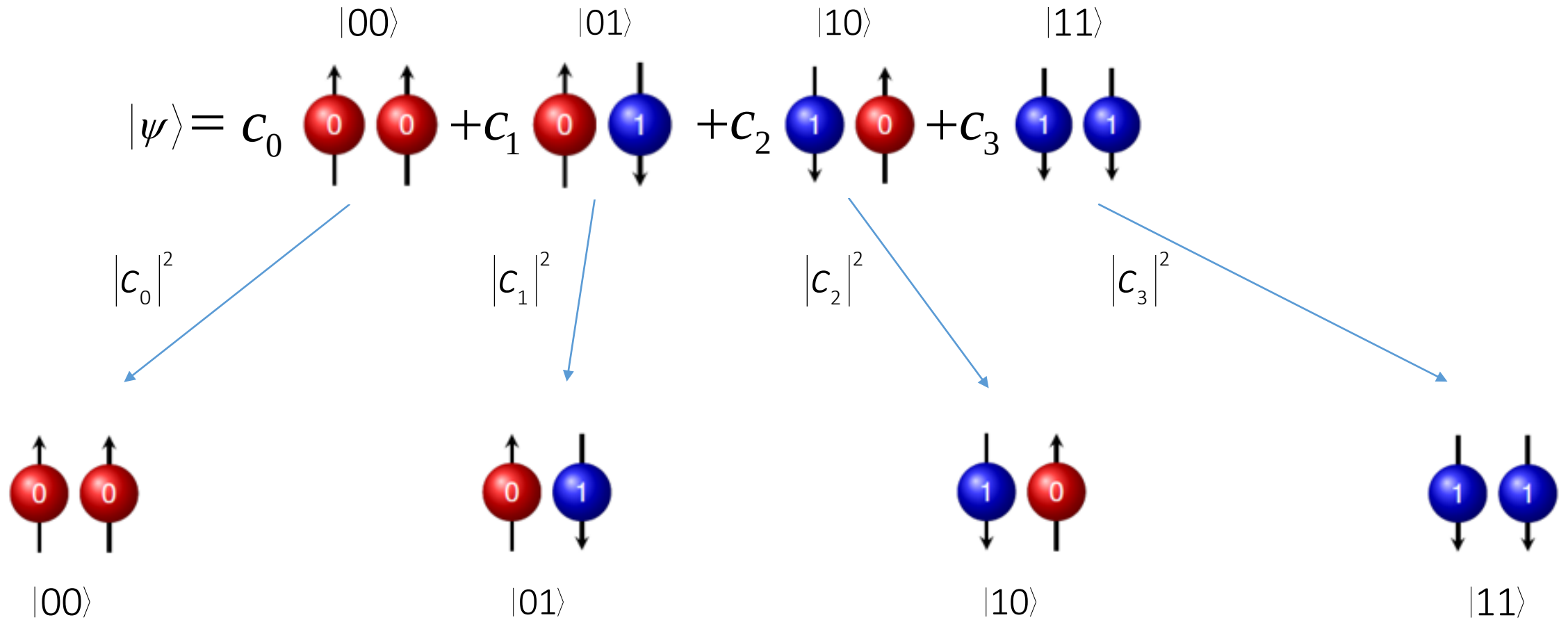
Notations

- An analogous notation is also used for operators on tensor product spaces

$$H^{\otimes k} = H \otimes H \otimes \dots \otimes H$$

where H is the Hadamard operator

Multiple Qubits Measurement



$$|c_0|^2 + |c_1|^2 + |c_2|^2 + |c_3|^2 = 1$$

Multiple Qubits Measurement (Example)

- Say we have two qubits in the state

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle + \frac{\sqrt{3}}{4}|10\rangle + \frac{1}{4}|11\rangle$$

- If we measured *both qubits*, we would get

$|00\rangle$ with probability $1/2$

$|01\rangle$ with probability $1/4$

$|10\rangle$ with probability $3/16$

$|11\rangle$ with probability $1/16$

Measuring Individual Qubits

- For a **two** qubit system, we could measure, for example, the first qubit

$$|\psi\rangle = c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle$$

- Measuring the **first** qubit alone gives 0 with probability

$$|c_0|^2 + |c_1|^2$$

leaving the post-measurement state

$$|\psi\rangle = \frac{c_0|00\rangle + c_1|01\rangle}{\sqrt{|c_0|^2 + |c_1|^2}}$$

- Note how the post-measurement state is **re-normalized**

Measuring Individual Qubits

- As an example, if

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle + \frac{\sqrt{3}}{4}|10\rangle + \frac{1}{4}|11\rangle$$

let us only measure the *left qubit*

Measuring Individual Qubits

- The probability of getting $|0\rangle$ when measuring the left qubit is given by the sum of the norm-squares of the amplitudes of $|00\rangle$ and $|01\rangle$, since those both have the left qubit as $|0\rangle$

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle + \frac{\sqrt{3}}{4}|10\rangle + \frac{1}{4}|11\rangle \quad \longrightarrow \quad \left|\frac{1}{\sqrt{2}}\right|^2 + \left|\frac{1}{2}\right|^2 = \frac{3}{4}$$

- Then the state collapses to the parts where the left qubit is $|0\rangle$, so it becomes A

$$|\psi_{|0\rangle}\rangle = A\left(\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle\right) \quad \longrightarrow \quad A = \frac{2}{\sqrt{3}} \quad \text{normalization constant}$$

Measuring Individual Qubits

- Similarly, if the outcome is $|1\rangle$, then from the $|10\rangle$ and $|11\rangle$ states, the probability is

$$|\psi\rangle = \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{2}|01\rangle + \frac{\sqrt{3}}{4}|10\rangle + \frac{1}{4}|11\rangle \quad \longrightarrow \quad \left|\frac{\sqrt{3}}{4}\right|^2 + \left|\frac{1}{4}\right|^2 = \frac{1}{4}$$

- Similarly, the results of the outcome is $|1\rangle$ then the state collapses to the terms where the left qubit is $|1\rangle$, so it becomes

$$|\psi_{|1\rangle}\rangle = B\left(\frac{\sqrt{3}}{4}|10\rangle + \frac{1}{4}|11\rangle\right) \quad \longrightarrow \quad B=2 \quad \text{normalization constant}$$

Product States

- Some quantum states can be factored into (the tensor product of) individual qubit states. For example,

$$\begin{aligned}\frac{1}{2}(|00\rangle - |01\rangle + |10\rangle - |11\rangle) &= \underbrace{\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)}_{|+\rangle} \otimes \underbrace{\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)}_{|-\rangle} \\ &= |+\rangle \otimes |-\rangle \\ &= |+\rangle|-\rangle.\end{aligned}$$

- Such factorizable states are called *product states* or simply *separable states*

Product States

Let us work through an example of how to factor a state. Say two qubits are the state

$$\frac{1}{2\sqrt{2}} \left(\sqrt{3}|00\rangle - \sqrt{3}|01\rangle + |10\rangle - |11\rangle \right).$$

We want to write this as the product of two single-qubit states,

$$|\psi_1\rangle|\psi_0\rangle,$$

where

$$|\psi_1\rangle = \alpha_1|0\rangle + \beta_1|1\rangle, \quad |\psi_0\rangle = \alpha_0|0\rangle + \beta_0|1\rangle.$$

Then,

$$\begin{aligned} |\psi_1\rangle|\psi_0\rangle &= (\alpha_1|0\rangle + \beta_1|1\rangle)(\alpha_0|0\rangle + \beta_0|1\rangle) \\ &= \alpha_1\alpha_0|00\rangle + \alpha_1\beta_0|01\rangle + \beta_1\alpha_0|10\rangle + \beta_1\beta_0|11\rangle. \end{aligned}$$

Matching up the coefficients with our original state,

$$\alpha_1 \alpha_0 = \frac{\sqrt{3}}{2\sqrt{2}}, \quad \alpha_1 \beta_0 = \frac{-\sqrt{3}}{2\sqrt{2}}, \quad \beta_1 \alpha_0 = \frac{1}{2\sqrt{2}}|10\rangle, \quad \beta_1 \beta_0 = \frac{-1}{2\sqrt{2}}.$$

$$\frac{1}{2\sqrt{2}} \left(\sqrt{3}|00\rangle - \sqrt{3}|01\rangle + |10\rangle - |11\rangle \right)$$

Using these equations, let us solve for the variables in terms of one of them. Starting with the first equation, we can solve for α_1 in terms of α_0 :

$$\alpha_1 \alpha_0 |00\rangle + \alpha_1 \beta_0 |01\rangle + \beta_1 \alpha_0 |10\rangle + \beta_1 \beta_0 |11\rangle$$

$$\alpha_1 = \frac{\sqrt{3}}{2\sqrt{2}\alpha_0}.$$

Plugging this into the second equation, we can solve for β_0 in terms of α_0 :

$$\beta_0 = -\alpha_0.$$

For the third equation, we can solve for β_1 in terms of α_0 :

$$\beta_1 = \frac{1}{2\sqrt{2}\alpha_0}.$$

Finally, plugging in $\beta_1 = \frac{1}{2\sqrt{2}} \frac{1}{\alpha_0}$ and $\beta_0 = -\alpha_0$ into the fourth equation, we get

$$\frac{-1}{2\sqrt{2}} = \frac{-1}{2\sqrt{2}},$$

which is a true statement, so it is satisfied, although it does not tell us anything new. So, we have solve for α_1 , β_1 , and β_0 in terms of α_0 , and this is actually sufficient.

Plugging into the product state,

$$\begin{aligned} |\psi_1\rangle|\psi_0\rangle &= (\alpha_1|0\rangle + \beta_1|1\rangle)(\alpha_0|0\rangle + \beta_0|1\rangle) \\ &= \left(\frac{\sqrt{3}}{2\sqrt{2}\alpha_0}|0\rangle + \frac{1}{2\sqrt{2}}\frac{1}{\alpha_0}|1\rangle \right) (\alpha_0|0\rangle - \alpha_0|1\rangle). \end{aligned}$$

We see that α_0 cancels, yielding

$$|\psi_1\rangle|\psi_0\rangle = \left(\frac{\sqrt{3}}{2\sqrt{2}}|0\rangle + \frac{1}{2\sqrt{2}}|1\rangle \right) (|0\rangle - |1\rangle).$$

Moving the factor of $1/\sqrt{2}$ to the right qubit so that both qubits are normalized,

$$|\psi_1\rangle|\psi_0\rangle = \left(\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle \right) \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right).$$

Thus, the left qubit is in the state $\frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$, and the right qubit is in the state $|-\rangle$.

In general, a product state of n qubits can be written

$$(\alpha_{n-1}|0\rangle + \beta_{n-1}|1\rangle) \otimes \cdots \otimes (\alpha_1|0\rangle + \beta_1|1\rangle) \otimes (\alpha_0|0\rangle + \beta_0|1\rangle).$$

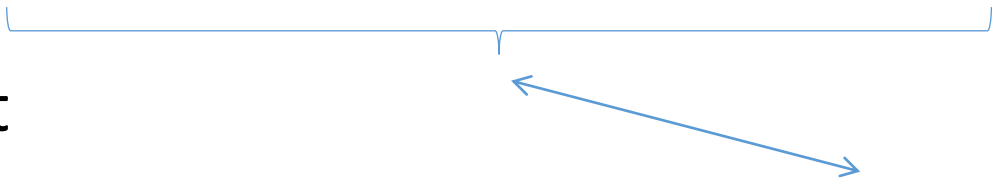
Entanglement

- There exist quantum states that *cannot be factored into product states*
- These are called *entangled states*
- The *entanglement* is one of the most interesting and puzzling ideas associated with composite quantum systems
- Consider the two qubit state named the *Bell State* or *EPR pair*

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \rightarrow \frac{1}{\sqrt{2}} \begin{array}{c} \uparrow \\ \text{0} \end{array} \begin{array}{c} \uparrow \\ \text{0} \end{array} + \frac{1}{\sqrt{2}} \begin{array}{c} \downarrow \\ \text{1} \end{array} \begin{array}{c} \downarrow \\ \text{1} \end{array}$$

Entanglement

- $|\Phi^+\rangle$ **cannot** be written as $|\psi_1\rangle|\psi_0\rangle$
- As a proof, let us try writing it in a product using the same procedure as in the previous slides

$$|\psi_1\rangle|\psi_0\rangle = (\alpha_1|0\rangle + \beta_1|1\rangle)(\alpha_0|0\rangle + \beta_0|1\rangle) = \alpha_1\alpha_0|00\rangle + \alpha_1\beta_0|01\rangle + \beta_1\alpha_0|10\rangle + \beta_1\beta_0|11\rangle$$


- Matching the coefficients, we get

$$\alpha_1\alpha_0 = \frac{1}{\sqrt{2}}, \quad \alpha_1\beta_0 = 0, \quad \beta_1\alpha_0 = 0, \quad \beta_1\beta_0 = \frac{1}{\sqrt{2}}$$

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

Entanglement

$$\alpha_1\alpha_0 = \frac{1}{\sqrt{2}}, \quad \alpha_1\beta_0 = 0, \quad \beta_1\alpha_0 = 0, \quad \beta_1\beta_0 = \frac{1}{\sqrt{2}}$$

- The second equation requires $\alpha_1 = 0$ or $\beta_0 = 0$
- If $\alpha_1 = 0$, then the *first equation* gives $0 = \frac{1}{\sqrt{2}}$, which is false
- If $\beta_0 = 0$, then the *fourth equation* gives $0 = \frac{1}{\sqrt{2}}$, which is false
- Thus, there is no solution to these four equations, so $|\Phi^+\rangle$ cannot be written as a product state
- Therefore, $|\Phi^+\rangle$ is an *entangled state*

Bell States or EPR Pair

*physicists
use this alternative symbols*

$$\begin{aligned} |\beta_{00}\rangle &= \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \rightarrow |\Phi^+\rangle \\ |\beta_{01}\rangle &= \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \rightarrow |\Psi^+\rangle \\ |\beta_{10}\rangle &= \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) \rightarrow |\Phi^-\rangle \\ |\beta_{11}\rangle &= \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \rightarrow |\Psi^-\rangle \end{aligned}$$

These are the **only** four **maximally** entangled pairs, and they are known as the *Bell states* or *EPR states* or *EPR pairs* (for the physicists **E**instein, **P**odolsky, and **R**osen)

Bell States or EPR Pair

- The mnemonic notation,

$$|\beta_{00}\rangle, |\beta_{01}\rangle, |\beta_{10}\rangle, |\beta_{11}\rangle$$

can be easily understood via the equations

$$|\beta_{xy}\rangle = \frac{1}{\sqrt{2}} \left(|0, y\rangle + (-1)^x |1, \bar{y}\rangle \right)$$

where $\bar{y}$ is the negation of y

Entanglement Measure

- Given the importance of entanglement in quantum computing, we need a way to quantify the *degree of entanglement within a state*, i.e., we need an *entanglement measure*, to characterize the degree of entanglement within a *2-qubit quantum state*
- There are many of such performance measures
- Fortunately, in the case of 2-qubit states, all the different entanglement measures turn out to be equivalent to one another
- However, no such equivalence is found for entanglement measures of n -qubit states

Entanglement Measure

- Specifically, the *tangle* provides a *quantitative measure* of *the degree of entanglement* within a quantum state
- Formally, the *tangle is the square of the concurrence*, which for a 2-qubit pure state, $|\psi\rangle$, is given by:

$$\text{Concurrence}|\psi\rangle = |\langle\psi|\tilde{\psi}\rangle|$$

where $|\tilde{\psi}\rangle$ is the spin-flipped version of $|\psi\rangle$ which is defined as

$$|\tilde{\psi}\rangle = (Y \otimes Y)|\psi^*\rangle$$

Entanglement Measure

- If

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle \longrightarrow |\psi^*\rangle = a^*|00\rangle + b^*|01\rangle + c^*|10\rangle + d^*|11\rangle$$

and therefore

$$|\tilde{\psi}\rangle = (Y \otimes Y)|\psi^*\rangle = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \otimes \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} a^* \\ b^* \\ c^* \\ d^* \end{bmatrix} = \begin{bmatrix} 0 \cdot \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} & -i \cdot \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \\ i \cdot \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} & 0 \cdot \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \end{bmatrix} \begin{bmatrix} a^* \\ b^* \\ c^* \\ d^* \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a^* \\ b^* \\ c^* \\ d^* \end{bmatrix} = \begin{bmatrix} -d^* \\ c^* \\ b^* \\ -a^* \end{bmatrix} = -d^*|00\rangle + c^*|01\rangle + b^*|10\rangle - a^*|11\rangle$$

Entanglement Measure

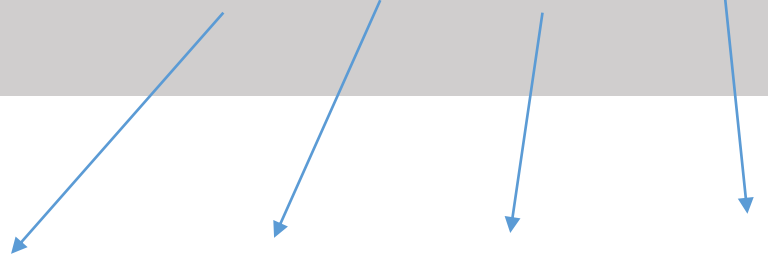
- Thus

$$|\tilde{\psi}\rangle = (Y \otimes Y)|\psi^*\rangle = -d^*|00\rangle + c^*|01\rangle + b^*|10\rangle - a^*|11\rangle$$

and therefore the concurrence of a general 2-qubit state $|\psi\rangle$ is given by:

$$\text{Concurrence}|\psi\rangle = |\langle\psi|\tilde{\psi}\rangle| = |2b^*c^* - 2a^*d^*| = 2|b^*c^* - a^*d^*|$$

Entanglement Measure

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$$


- If $|\psi\rangle$ is *separable*

$$|\psi\rangle = |\psi_1\rangle|\psi_0\rangle = (\alpha_1|0\rangle + \beta_1|1\rangle)(\alpha_0|0\rangle + \beta_0|1\rangle) = \alpha_1\alpha_0|00\rangle + \alpha_1\beta_0|01\rangle + \beta_1\alpha_0|10\rangle + \beta_1\beta_0|11\rangle$$

then

$$a = \alpha_1\alpha_0 \quad b = \alpha_1\beta_0 \quad c = \beta_1\alpha_0 \quad d = \beta_1\beta_0$$

- Hence, if a, b, c and d are different from zero, the *concurrence* of a general 2-qubit state $|\psi\rangle$ is given by:

$$\text{Concurrence}|\psi\rangle = |\langle\psi|\tilde{\psi}\rangle| = |2b^*c^* - 2a^*d^*| = 0$$



Unentangled states have a concurrence of zero

Entanglement Measure

- At the other extreme, under the spin-flip transformation *maximally entangled states*, such as Bell states, remain invariant up to an unimportant overall phase
- To see this, the four Bell states are given by

$$\begin{array}{lcl} |\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) & & -|\beta_{00}\rangle \\ |\beta_{01}\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) & \xrightarrow{\quad (Y \otimes Y) |\beta_{xy}^*\rangle \quad} & +|\beta_{01}\rangle \\ |\beta_{10}\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) & & +|\beta_{10}\rangle \\ |\beta_{11}\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) & & -|\beta_{11}\rangle \end{array} \quad \Rightarrow \quad \text{Concurrence } |\beta_{xy}\rangle = \left| \langle \beta_{xy} | \beta_{xy} \rangle \right| = 1$$

Entanglement Measure

- Hence, the overlap between a maximally entangled state and its spin-flipped counterpart is unity, which is the most it can be, implying that maximally entangled states have a concurrence of one
- Thus the *tangle*, as defined above, provides a quantitative measure for the *degree of entanglement* within a pure 2-qubit state
- Generalizations of tangle to mixed states and multi-partite states are beyond the scope of these seminars

Entanglement Measure

- Example

$$|\psi\rangle = \sqrt{\frac{1}{3}}|00\rangle + \sqrt{\frac{2}{3}}|11\rangle \quad \longrightarrow \quad \begin{cases} a = a^* = \sqrt{\frac{1}{3}} \\ b = b^* = 0 \\ c = c^* = 0 \\ d = d^* = \sqrt{\frac{2}{3}} \end{cases}$$

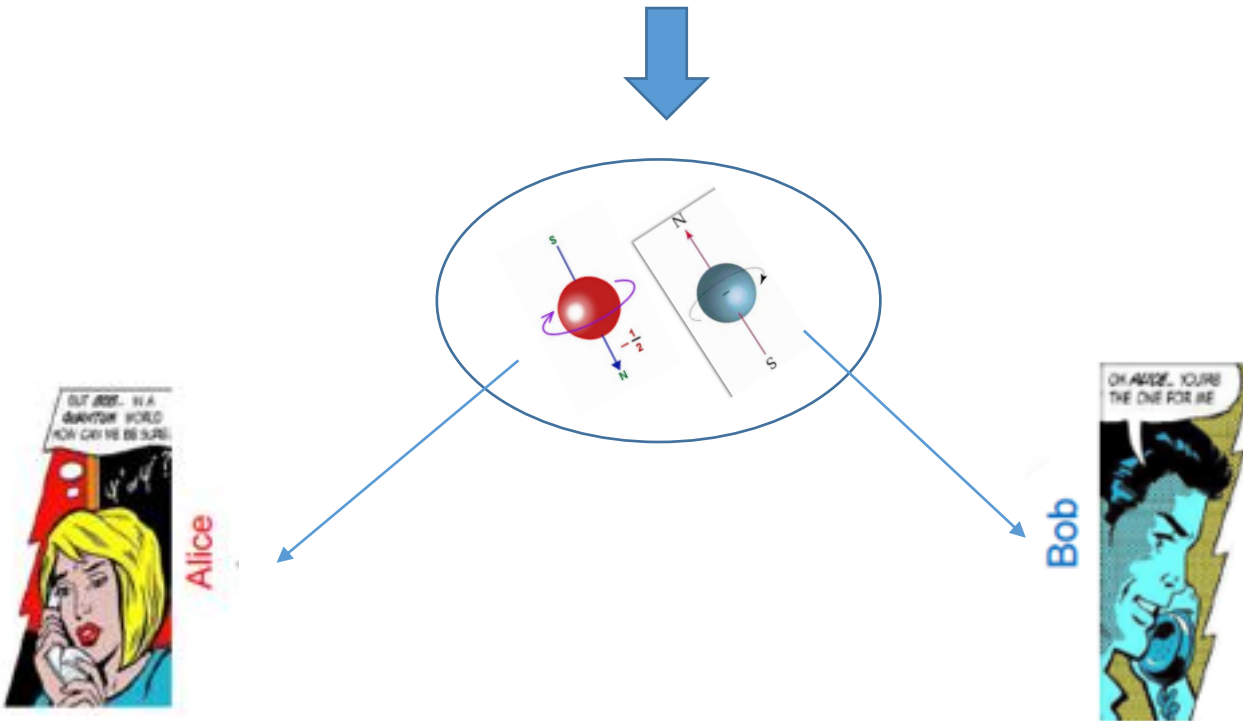
- Thus

$$\text{Concurrence}|\psi\rangle = |\langle\psi|\tilde{\psi}\rangle| = |2b^*c^* - 2a^*d^*| = \left|0 - 2\frac{\sqrt{2}}{3}\right| \cong 0.93$$

Entanglement

qubits *A* and *B*

$$|\Psi\rangle = \frac{1}{\sqrt{2}} [|0_A 0_B\rangle + |1_A 1_B\rangle]$$



Entanglement

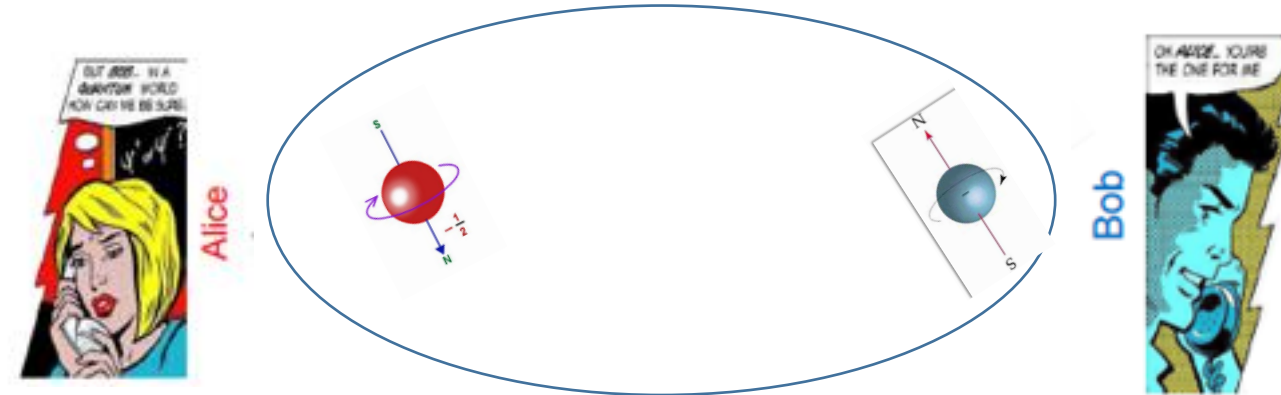
- Upon measuring the first qubit, Alice obtains two possible results

→ 0 with probability $\frac{1}{2}$, leaving the post-measurement state $|\varphi\rangle = |0_A 0_B\rangle$

→ 1 with probability $\frac{1}{2}$, leaving the post-measurement state $|\psi\rangle = |1_A 1_B\rangle$

qubits *A* and *B*

$$|\Psi\rangle = \frac{1}{\sqrt{2}} [|0_A 0_B\rangle + |1_A 1_B\rangle]$$



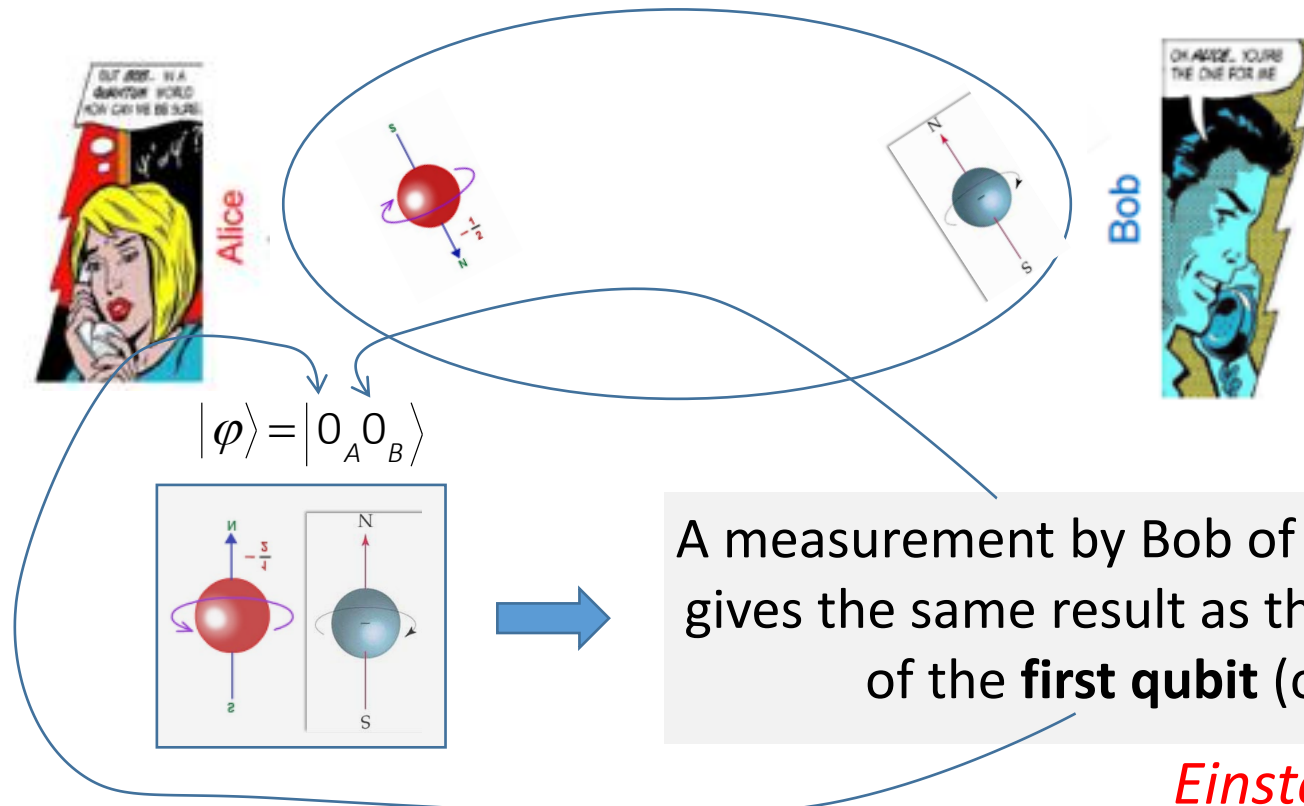
$$|\varphi\rangle = |0_A 0_B\rangle$$

Assume that the Alice outcome is 0

Entanglement

qubits **A** and **B**

$$|\Psi\rangle = \frac{1}{\sqrt{2}} [|0_A 0_B\rangle + |1_A 1_B\rangle]$$



Einstein's "Spooky Action at a Distance"

Non-Locality

- Entangled states are also said to be *non-local*, meaning that if you are in a room with only one of the two qubit you do not have *full control* over what happens to the data there; the owner of the other qubit in a different room may measure his qubit and affect your data (i.e. qubit state) even though you have done nothing
- Furthermore, if you measure the state of your qubit, your efforts are not confined to your room but extend to the outside world where the other qubit is located

Locality

- Likewise, separable states are considered *local*, since they do allow full segregation of the actions on separate qubit states
- Each observer has total control of the destiny of his qubit, and his actions does not affect the other observer

A Quantum Model of Computation

The Quantum Circuit Model

- We've already met a few simple quantum circuits
- Let's look in a little more detail at the elements of a quantum circuit
- A simple quantum circuit containing three quantum gates has already been discussed
- The output of the third gate (Z) is $ZSH|0\rangle$



The Quantum Circuit Model

- Each line in the circuit represents a *wire* in the quantum circuit
- This wire does not necessarily correspond to a physical wire; it may correspond instead to a physical particle such as a photon moving from one location to another through space
- It is conventional to assume that the state input to the circuit is a computational basis state, *usually* the state consisting of $|0\rangle$

However, other input states, such as $|+\rangle, |-\rangle$, are possible

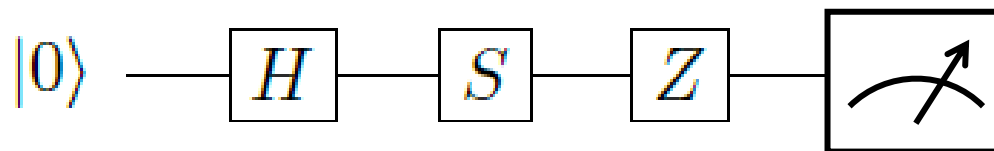


Combination of Quantum Gates

- The circuit is read left-to-right, just like a classical circuit diagram
- So we start with a single qubit in the $|0\rangle$ state and apply a Hadamard gate H to it, followed by a phase gate S , and finally a Z gate

$$ZSH|0\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \frac{1}{\sqrt{2}} [|0\rangle - i|1\rangle]$$

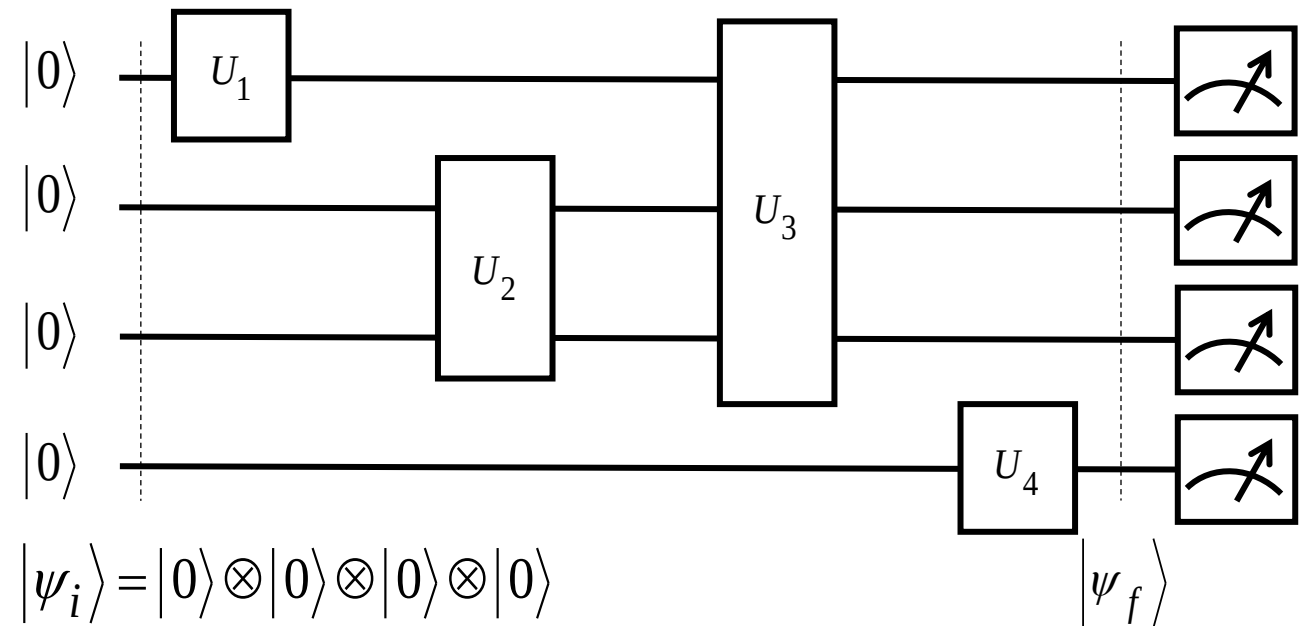
and if we *measure* the qubit we get $|0\rangle$ or $|1\rangle$ with equal probability



The Quantum Circuit Model

As we proceed we'll introduce new quantum gates as needed.

The previous quantum circuit model can be extended to quantum circuit models that are capable of transforming the state of two qubits or more.



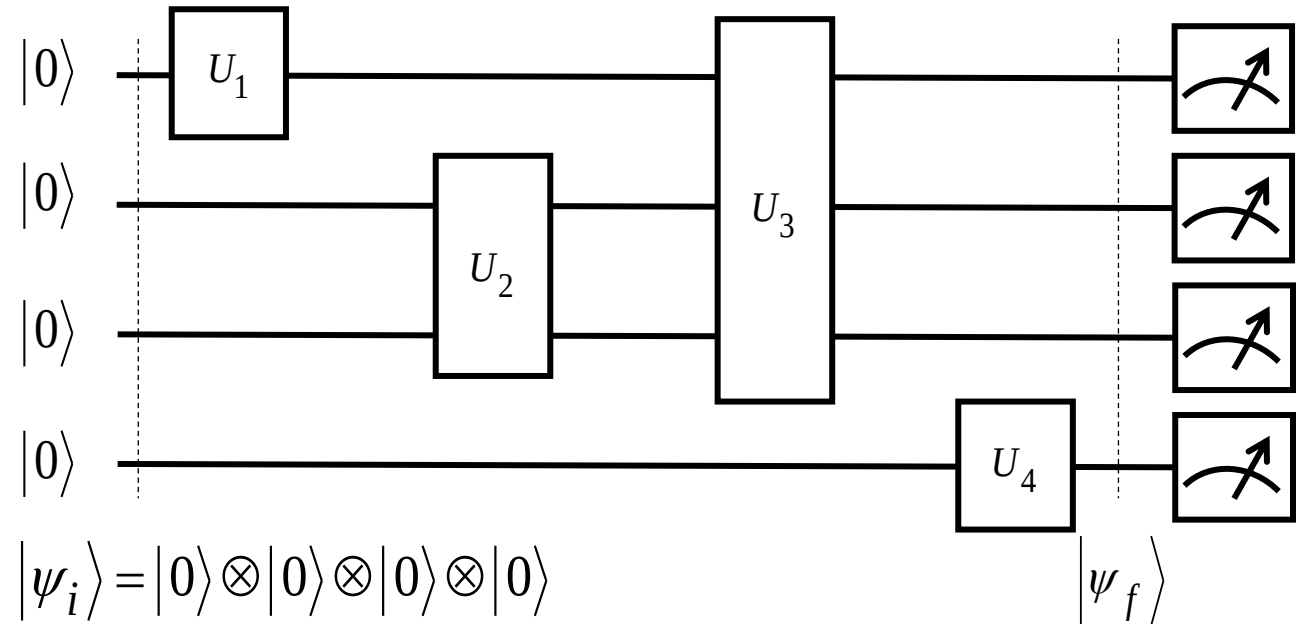
The Quantum Circuit Model

The four qubit state

$$|\psi_i\rangle = |0\rangle \otimes |0\rangle \otimes |0\rangle \otimes |0\rangle$$

enters the circuit at the left.

These qubits are processed by the gates U_1 , U_2 , U_3 , and U_4 .

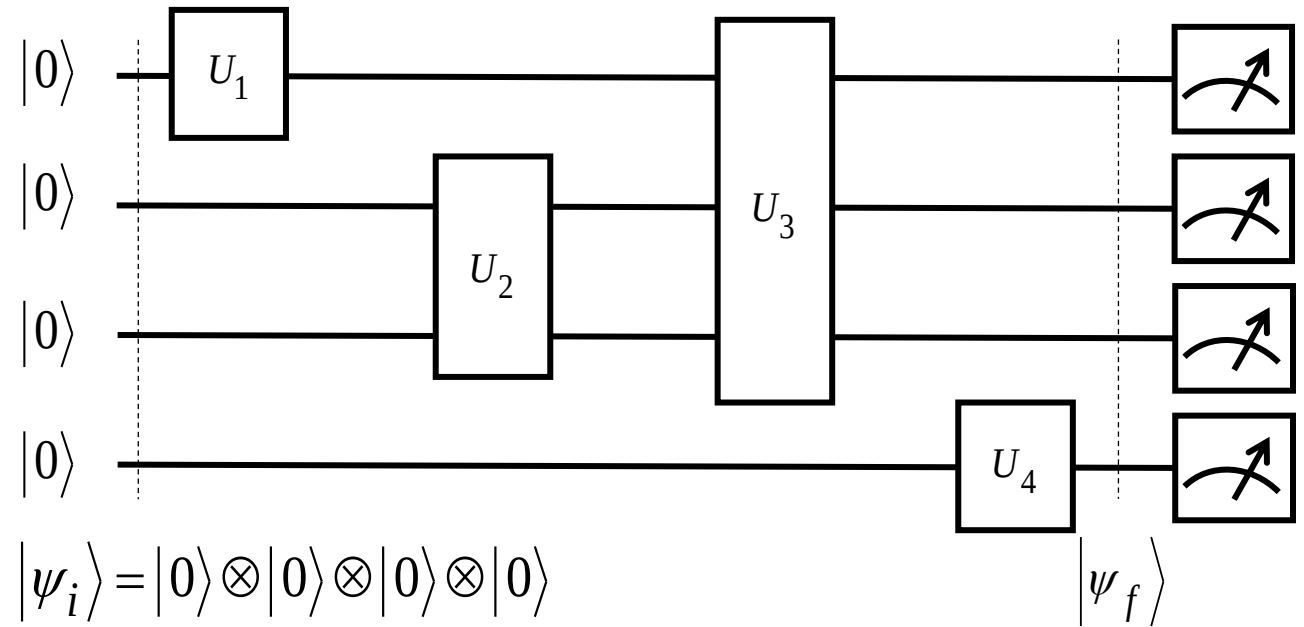


At the output of the circuit we have the collective 4-qubit state $|\psi_f\rangle$

The Quantum Circuit Model

A measurement is then made of the resulting state

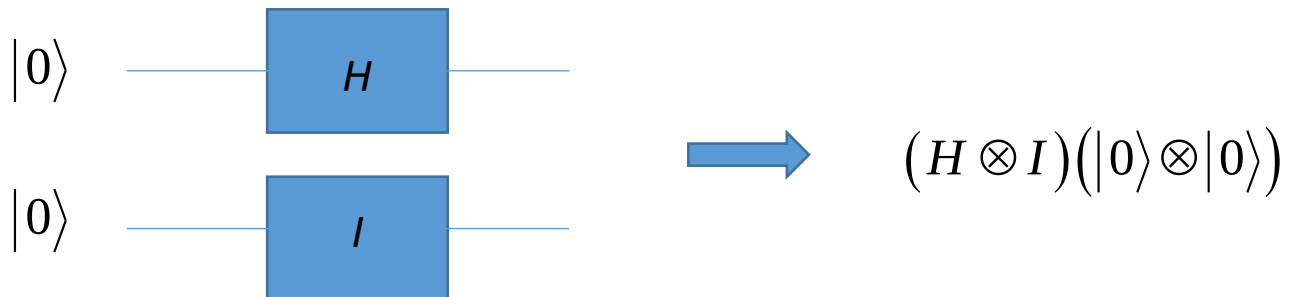
The measurement will often be a simple qubit-by-qubit measurement in the CBS basis, but in some cases may be a more general measurement of the joint state.



In the following, we take a closer look at the two-qubit gates

Two-Qubit Quantum Gates

- Assume we have multiple qubits, and we want to apply a single-qubit gate (like I , X , Y , Z , S , T , or H) to just a single qubit
- For example, say we have two qubits in the $|00\rangle = |0\rangle \otimes |0\rangle$ state, and we want to apply the Hadamard gate to the left qubit, but leave the right qubit alone (i.e., apply the identity gate to it)
- We write the gates using a tensor product, so we write

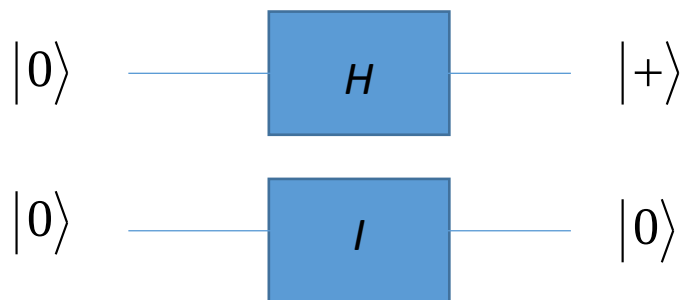


Two-Qubit Quantum Gates

$$\begin{aligned}
 (H \otimes I)(|0\rangle \otimes |0\rangle) &= H|0\rangle \otimes I|0\rangle = |+\rangle \otimes |0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle \\
 &= \frac{1}{\sqrt{2}}(|0\rangle \otimes |0\rangle + |1\rangle \otimes |0\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}
 \end{aligned}$$

$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$|10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$



$$(H \otimes I)(|0\rangle \otimes |0\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Two-Qubit Quantum Gates

- We can find $H \otimes I$ as a matrix in two different ways
- **First**, we can find how $H \otimes I$ acts on each of the basis states $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$
- We already found how it acts on $|00\rangle$. We just carry on with the rest.

$$(H \otimes I)|00\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$(H \otimes I)|01\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

$$(H \otimes I)|10\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$$

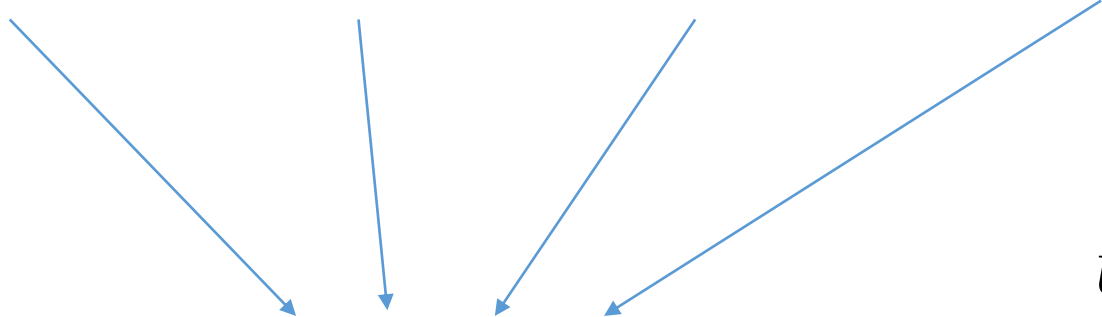
$$(H \otimes I)|11\rangle = \frac{1}{\sqrt{2}}(|01\rangle - |11\rangle) = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}$$

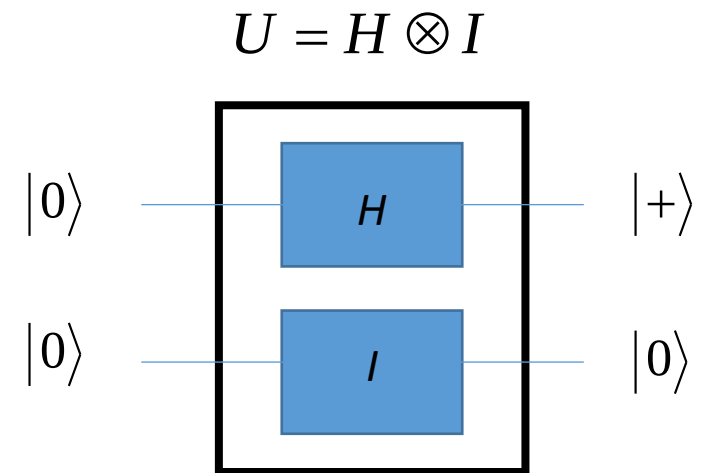
Two-Qubit Quantum Gates

- We can write $H \otimes I$ as a matrix by combining the column vectors for

$$H \otimes I \equiv [(H \otimes I)|00\rangle, (H \otimes I)|01\rangle, (H \otimes I)|10\rangle, (H \otimes I)|11\rangle]$$

as a 4x4 grid:


$$H \otimes I = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix}$$



Two-Qubit Quantum Gates

- The *second way* to find this matrix is by taking the Kronecker product of H and X :

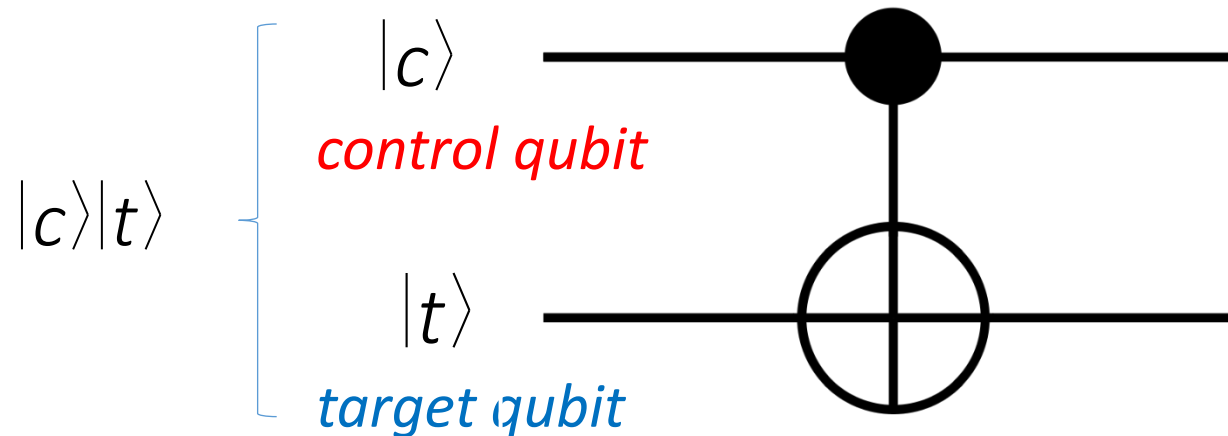
$$\begin{aligned} H \otimes I &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ 1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} & -1 \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \end{bmatrix} \end{aligned}$$

Two-Qubit Quantum Gates

- Quantum gates can also operate on two qubits at the same time
- Some important examples include:
 1. The *CNOT gate* or *controlled-NOT gate*, or since the X gate is the *NOT* gate, the *CNOT gate* is also called the *cX gate* or *controlled-X gate*
 2. The controlled- U gate
 3. The *SWAP* gate

CNOT Gate

- This is the simplest and most commonly used *2-qubit gate* in quantum computing
- The *controlled-NOT* or CNOT gate has two input qubits, known as the *control qubit* (top line) and the *target qubit* (bottom line), respectively

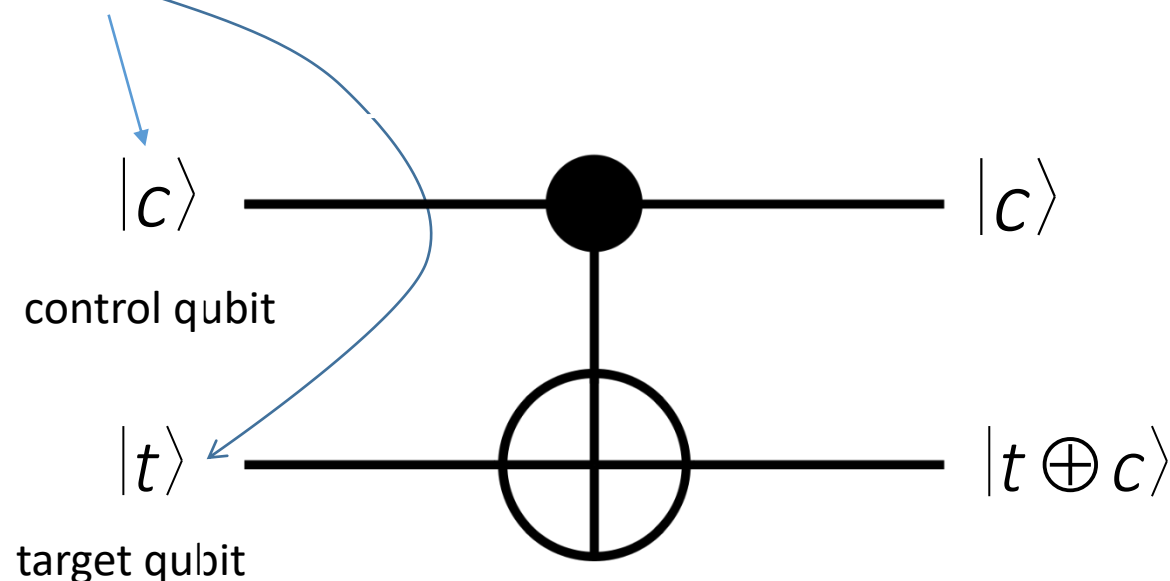


CNOT Gate

The action of the gate

- If the *control qubit* is set to $|0\rangle$, then *nothing happens to the target qubit*
- If the *control qubit* is set to $|1\rangle$, then *the target qubit is flipped*

$$|00\rangle \rightarrow |00\rangle; |01\rangle \rightarrow |01\rangle; |10\rangle \rightarrow |11\rangle; |11\rangle \rightarrow |10\rangle$$



The solid dot indicates control, and the $\oplus$ denotes the target, which is the XOR of the control and the target

CNOT Gate

- Formally

$$\begin{aligned} CNOT|00\rangle &= |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} & CNOT|01\rangle &= |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} & CNOT|10\rangle &= |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} & CNOT|11\rangle &= |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \end{aligned}$$

- Thus, in the computational basis $|control, target\rangle$ the matrix representation of CNOT is

$$CNOT = \begin{bmatrix} CNOT|00\rangle & CNOT|01\rangle & CNOT|10\rangle & CNOT|11\rangle \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

CNOT Gate

- Since

$$CNOT = CNOT^\dagger$$

and because

$$CNOT \cdot CNOT^\dagger = CNOT^\dagger \cdot CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I$$

it follows

$$CNOT^2 = I$$

CNOT Gate

- We can express *CNOT* by means of *outer products*

$$CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 11| + |11\rangle\langle 10|$$

CNOT Gate

$$(A \otimes B) \cdot (C \otimes D) = (AC) \otimes (BD)$$

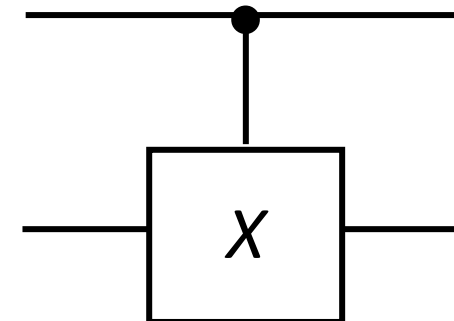
$$\begin{aligned} |00\rangle\langle 00| &= (|0\rangle_A \otimes |0\rangle_B)(\langle 0|_A \otimes \langle 0|_B) = (|0\rangle_A \langle 0|) \otimes (|0\rangle_B \langle 0|) & |10\rangle\langle 11| &= (|1\rangle_A \otimes |0\rangle_B)(\langle 1|_A \otimes \langle 1|_B) = (|1\rangle_A \langle 1|) \otimes (|0\rangle_B \langle 1|) \\ |01\rangle\langle 01| &= (|0\rangle_A \otimes |1\rangle_B)(\langle 0|_A \otimes \langle 1|_B) = (|0\rangle_A \langle 0|) \otimes (|1\rangle_B \langle 1|) & |11\rangle\langle 10| &= (|1\rangle_A \otimes |1\rangle_B)(\langle 1|_A \otimes \langle 0|_B) = (|1\rangle_A \langle 1|) \otimes (|1\rangle_B \langle 0|) \end{aligned}$$

- By exploiting the previous results

$$\begin{aligned} CNOT &= |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 11| + |11\rangle\langle 10| \\ &= (|0\rangle_A \langle 0|) \otimes (|0\rangle_B \langle 0|) + (|0\rangle_A \langle 0|) \otimes (|1\rangle_B \langle 1|) + (|1\rangle_A \langle 1|) \otimes (|0\rangle_B \langle 1|) + (|1\rangle_A \langle 1|) \otimes (|1\rangle_B \langle 0|) \\ &= |0\rangle_A \langle 0| \otimes (|0\rangle_B \langle 0| + |1\rangle_B \langle 1|) + |1\rangle_A \langle 1| \otimes (|0\rangle_B \langle 1| + |1\rangle_B \langle 0|) \\ &= |0\rangle_A \langle 0| \otimes I_B + |1\rangle_A \langle 1| \otimes X_B \end{aligned}$$



$$CNOT = |0\rangle_A \langle 0| \otimes I_B + |1\rangle_A \langle 1| \otimes X_B$$



Behavior on General State

- Applying CNOT to the general state

$$|\psi\rangle = c_0|00\rangle + c_1|01\rangle + c_2|10\rangle + c_3|11\rangle = \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix}$$

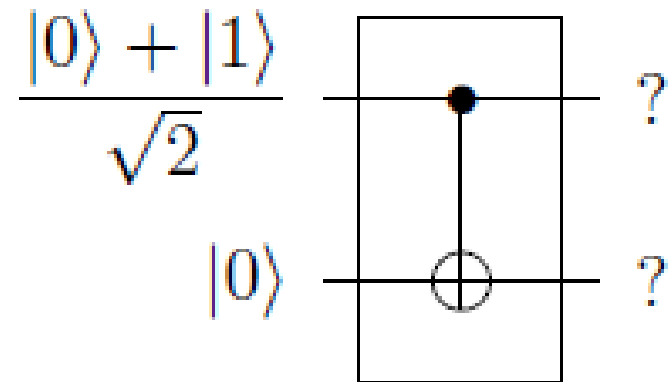
we get

$$CNOT|\psi\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} c_0 \\ c_1 \\ c_3 \\ c_2 \end{bmatrix} = c_0|00\rangle + c_1|01\rangle + c_3|10\rangle + c_2|11\rangle$$

Quantum Entanglement for CNOT

- A separable bipartite state into CNOT gate *does not usually result* in a *separable state* out of CNOT
- To see this, consider the separable state

$$|\psi\rangle = \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) |0\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |10\rangle)$$

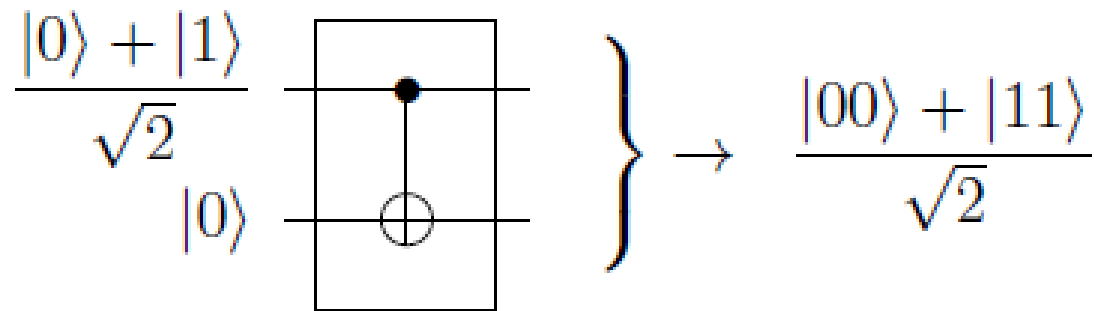


Quantum Entanglement for CNOT

- Now, apply CNOT using linearity,

$$CNOT|\psi\rangle = \frac{1}{\sqrt{2}}(CNOT|00\rangle + CNOT|10\rangle) = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

- This is the true output of the gate for the presented input
- The output state is not separable



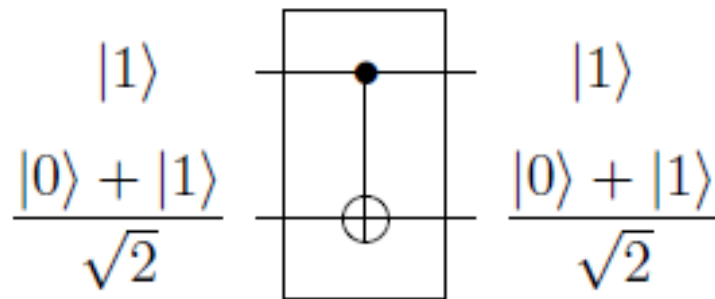
Quantum Entanglement for CNOT

- Now, consider the separable state

$$|\psi\rangle = |1\rangle \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}} \right) = \frac{1}{\sqrt{2}} (|10\rangle + |11\rangle)$$

and apply CNOT

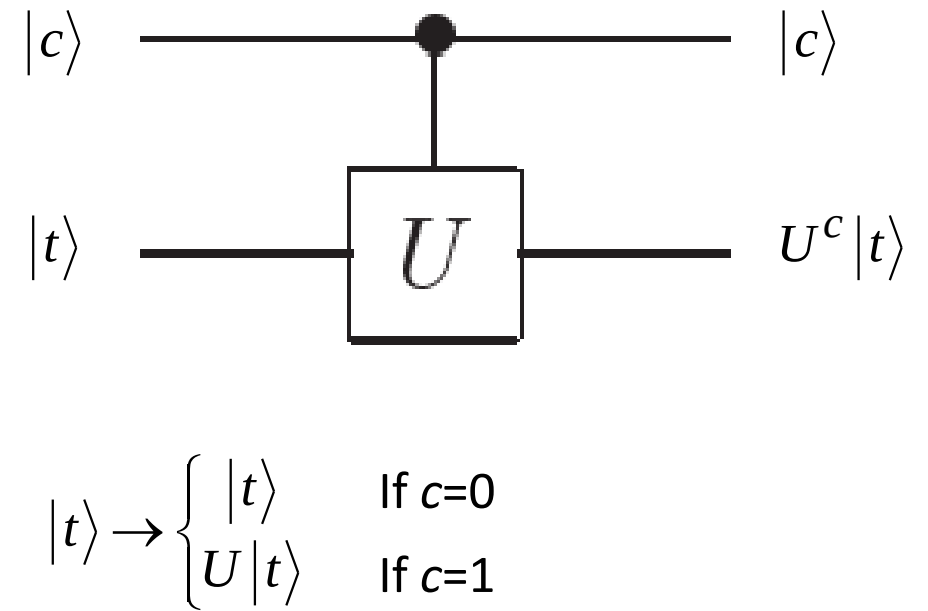
$$CNOT|\psi\rangle = \frac{1}{\sqrt{2}} (CNOT|10\rangle + CNOT|11\rangle) = \frac{1}{\sqrt{2}} (|11\rangle + |10\rangle) = |1\rangle \left(\frac{|1\rangle + |0\rangle}{\sqrt{2}} \right)$$



Aha! *separable*. That's because the control-bit is a CBS; it does not change during the linear application of CNOT so will be conveniently available for CNOT factoring at the output

The *controlled-U* Gate

- Suppose U is an arbitrary single qubit unitary operation
- A *controlled-U* (CU) operation is a two qubit operation, again with a control and a target qubit
- If the control qubit is set then U is applied to the target qubit, otherwise the target qubit is left alone; that is, $|c\rangle|t\rangle \rightarrow |c\rangle U^c |t\rangle$



The *controlled-U* Gate

- Formally

$$CU|00\rangle = |00\rangle$$

$$CU|01\rangle = |01\rangle$$

$$CU|10\rangle = |1\rangle \otimes U|0\rangle$$

$$CU|11\rangle = |1\rangle \otimes U|1\rangle$$

- To get the matrix representation of CU , first say U acts on a single qubit as

$$U|0\rangle = \begin{bmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} U_{00} \\ U_{10} \end{bmatrix} = U_{00}|0\rangle + U_{10}|1\rangle$$

$$U|1\rangle = \begin{bmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} U_{01} \\ U_{11} \end{bmatrix} = U_{01}|0\rangle + U_{11}|1\rangle$$

The *controlled-U* Gate

- Thus

$$CU|10\rangle = |1\rangle \otimes U|0\rangle = |1\rangle \otimes (U_{00}|0\rangle + U_{10}|1\rangle) = U_{00}|10\rangle + U_{10}|11\rangle$$

$$CU|11\rangle = |1\rangle \otimes U|1\rangle = |1\rangle \otimes (U_{01}|0\rangle + U_{11}|1\rangle) = U_{01}|10\rangle + U_{11}|11\rangle$$


$$U|0\rangle = \begin{bmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} U_{00} \\ U_{10} \end{bmatrix} = U_{00}|0\rangle + U_{10}|1\rangle$$
$$U|1\rangle = \begin{bmatrix} U_{00} & U_{01} \\ U_{10} & U_{11} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} U_{01} \\ U_{11} \end{bmatrix} = U_{01}|0\rangle + U_{11}|1\rangle$$

- Since

$$|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

To help understand the algebraic passages, I show the results obtained in the previous slide.

The *controlled-U* gate

- It follows

$$CU|00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad CU|01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad CU|10\rangle = \begin{bmatrix} 0 \\ 0 \\ U_{00} \\ U_{10} \end{bmatrix} \quad CU|11\rangle = \begin{bmatrix} 0 \\ 0 \\ U_{01} \\ U_{11} \end{bmatrix}$$

- Thus, in the computational basis the matrix representation of CU is

$$CU = [CU|00\rangle \quad CU|01\rangle \quad CU|10\rangle \quad CU|11\rangle] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix}$$

The *controlled-U* gate

Behavior on general state

$$\begin{aligned} |\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle &= \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} \quad \longrightarrow \quad CU|\psi\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ U_{00}\gamma + U_{01}\delta \\ U_{10}\gamma + U_{11}\delta \end{bmatrix} \\ &= \alpha|0\rangle|0\rangle + \beta|0\rangle|1\rangle + (U_{00}\gamma + U_{01}\delta)|1\rangle|0\rangle + (U_{10}\gamma + U_{11}\delta)|1\rangle|1\rangle \end{aligned}$$

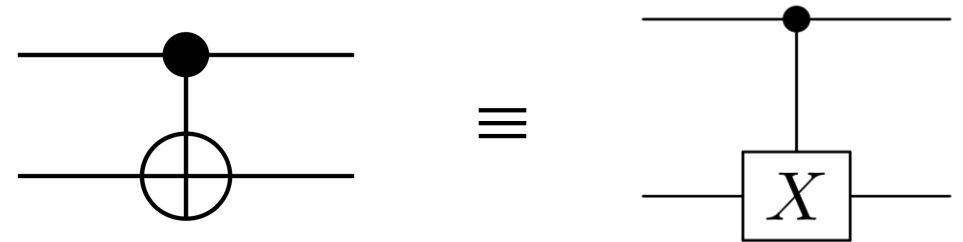
The *controlled-X* Gate

- Let's focus on the *controlled-X* gate
- In this case $U=X$

$$U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \begin{matrix} U_{00} = U_{11} = 0 \\ U_{01} = U_{10} = 1 \end{matrix} \quad \longrightarrow \quad CU = CX = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \text{CNOT}$$

CX can also be written as

$$CX = CNOT = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 11| + |11\rangle\langle 10|$$



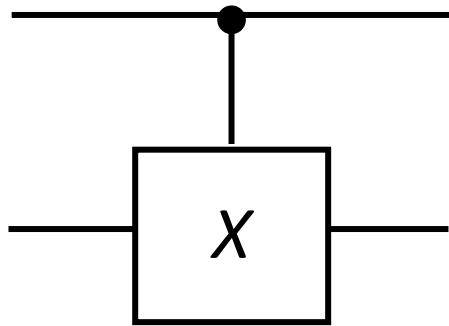
The *controlled-X* Gate

Behavior on general state

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix}$$



$$CX|\psi\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \delta \\ \gamma \end{bmatrix}$$



$$= \alpha|0\rangle|0\rangle + \beta|0\rangle|1\rangle + \delta|1\rangle|0\rangle + \gamma|1\rangle|1\rangle$$

The *controlled-Z* Gate

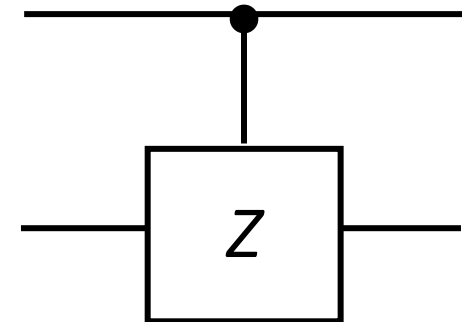
- Let's consider the *controlled-Z* gate
- In this case $U=Z$

$$U = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad \begin{matrix} U_{00} = 1 \\ U_{11} = -1 \\ U_{01} = U_{10} = 0 \end{matrix} \quad \longrightarrow$$

$$CU = CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$$

CZ can also be written as

$$CZ = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| - |11\rangle\langle 11|$$

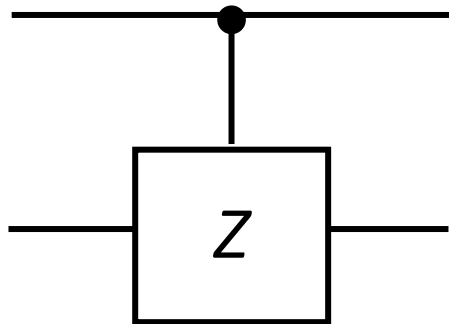


The *controlled*-Z Gate

Behavior on general state

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} \quad \rightarrow$$

$$CZ|\psi\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ -\delta \end{bmatrix}$$



$$= \alpha|0\rangle|0\rangle + \beta|0\rangle|1\rangle + \gamma|1\rangle|0\rangle - \delta|1\rangle|1\rangle$$

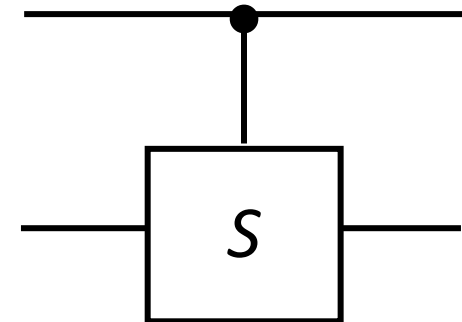
The *controlled-S* Gate

- Let's consider the controlled-S gate
- In this case $U=S$

$$U = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \quad \begin{matrix} U_{00} = 1 \\ U_{11} = i \\ U_{01} = U_{10} = 0 \end{matrix} \quad \longrightarrow \quad CU = CS = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & i \end{bmatrix}$$

CS can also be written as

$$CS = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + i|11\rangle\langle 11|$$



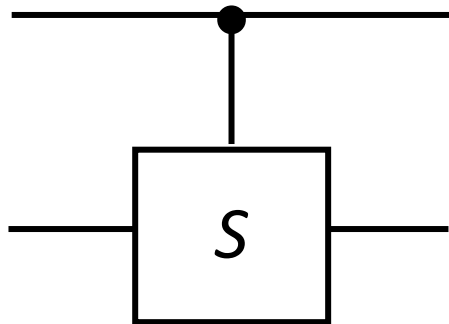
The *controlled-S* Gate

Behavior on general state

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix}$$



$$CS|\psi\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & i \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ i\delta \end{bmatrix}$$



$$= \alpha|0\rangle|0\rangle + \beta|0\rangle|1\rangle + \gamma|1\rangle|0\rangle + i\delta|1\rangle|1\rangle$$

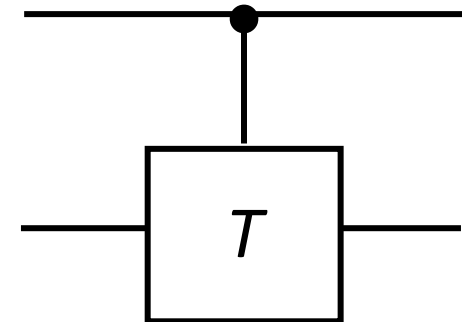
The *controlled-T* Gate

- Let's consider the controlled- T gate
- In this case $U=T$

$$U = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \quad \begin{matrix} U_{00} = 1 \\ U_{11} = e^{i\pi/4} \\ U_{01} = U_{10} = 0 \end{matrix} \quad \longrightarrow \quad CU = CT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & U_{00} & U_{01} \\ 0 & 0 & U_{10} & U_{11} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\pi/4} \end{bmatrix}$$

CT can also be written as

$$CT = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + e^{i\pi/4} |11\rangle\langle 11|$$



The *controlled-T* Gate

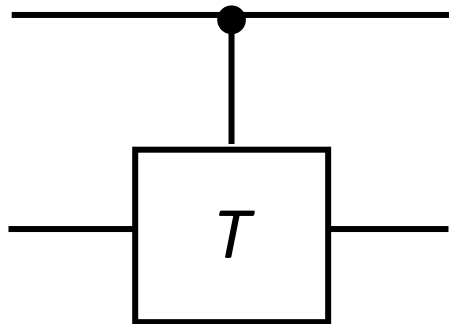
Behavior on general state

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix}$$



$$CT|\psi\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ e^{i\pi/4}\delta \end{bmatrix}$$

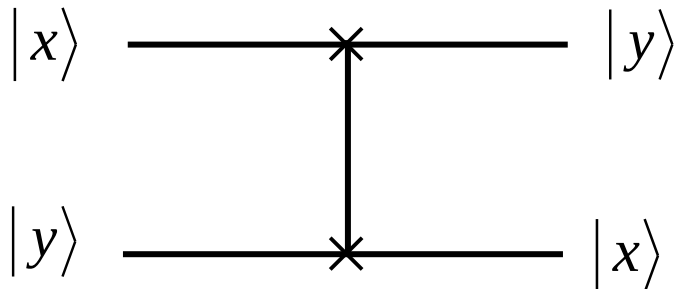
$$= \alpha|0\rangle|0\rangle + \beta|0\rangle|1\rangle + \gamma|1\rangle|0\rangle + e^{i\pi/4}\delta|1\rangle|1\rangle$$



The *SWAP* Gate

- The *SWAP* gate is two-qubit operation
- Expressed in computational basis, the *SWAP* gate swaps the state of the two qubits involved in the operation

$$\text{SWAP}|00\rangle = |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad \text{SWAP}|01\rangle = |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \quad \text{SWAP}|10\rangle = |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} \quad \text{SWAP}|11\rangle = |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$



In other words, $\text{SWAP}|x\rangle|y\rangle = |y\rangle|x\rangle$ for $x, y \in \{0, 1\}$

The *SWAP* Gate

- Thus, in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ the matrix representation of SWAP is

$$SWAP = \begin{bmatrix} SWAP|00\rangle & SWAP|01\rangle & SWAP|10\rangle & SWAP|11\rangle \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- *SWAP* can also be written as

$$SWAP = |00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01| + |11\rangle\langle 11|$$

The *SWAP* Gate

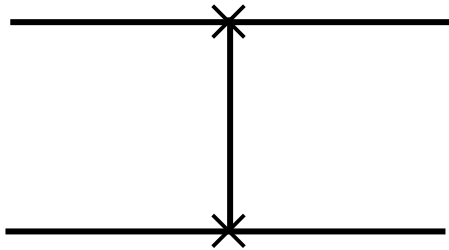
Behavior on general state

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle = \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix}$$



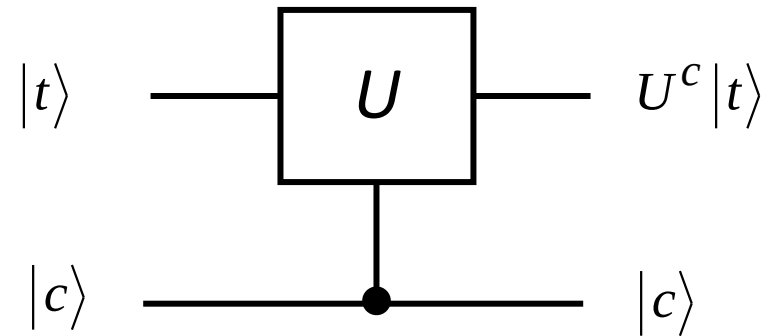
$$SWAP|\psi\rangle = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} \alpha \\ \gamma \\ \beta \\ \delta \end{bmatrix}$$

$$= \alpha|0\rangle|0\rangle + \gamma|0\rangle|1\rangle + \beta|1\rangle|0\rangle + \delta|1\rangle|1\rangle$$



Swapping Roles in *controlled-U* Gate

- We could have (and still can) turn any of our controlled gates upside down
- Let's refer to this version of a controlled gate using the notation $(C \uparrow)U$
- It is easy to derive the matrix on the CBS basis



$$|t\rangle \rightarrow \begin{cases} |t\rangle & \text{If } c=0 \\ U|t\rangle & \text{If } c=1 \end{cases}$$

Swapping Roles in *controlled-U* Gate

- Thus

$$(c \uparrow)U|00\rangle = |00\rangle$$

$$(c \uparrow)U|10\rangle = |10\rangle$$

$$(c \uparrow)U|01\rangle = (U|0\rangle) \otimes |1\rangle = (U_{00}|0\rangle + U_{10}|1\rangle) \otimes |1\rangle = U_{00}|01\rangle + U_{10}|11\rangle$$

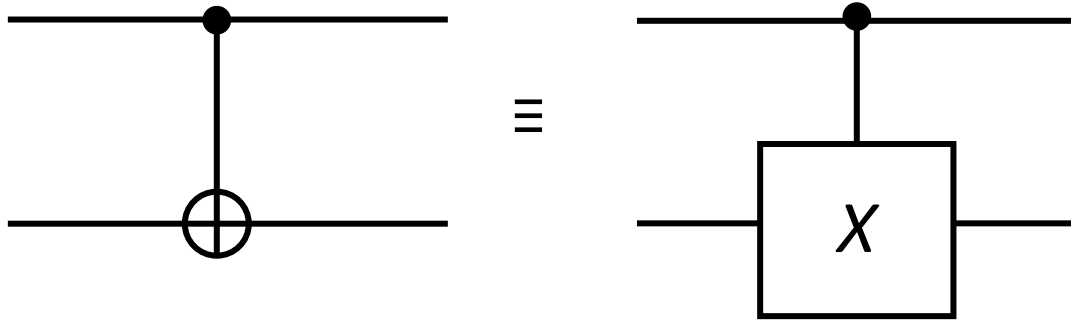
$$(c \uparrow)U|11\rangle = (U|1\rangle) \otimes |1\rangle = (U_{01}|0\rangle + U_{11}|1\rangle) \otimes |1\rangle = U_{01}|01\rangle + U_{11}|11\rangle$$

and the matrix we obtain is

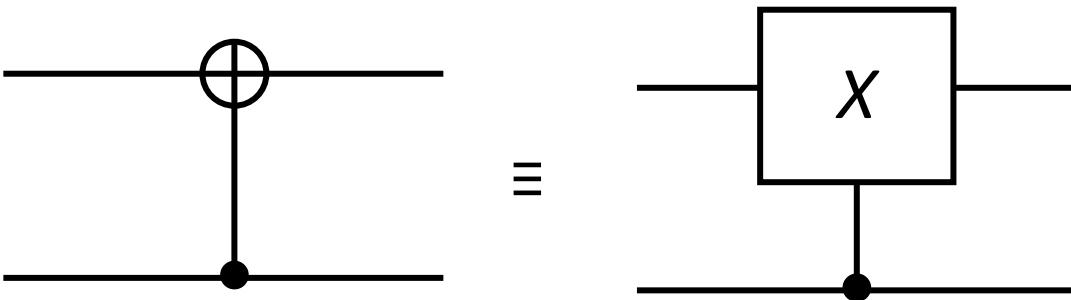
$$(c \uparrow)U = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & U_{00} & 0 & U_{01} \\ 0 & 0 & 1 & 0 \\ 0 & U_{10} & 0 & U_{11} \end{bmatrix}$$

Swapping Roles in *controlled-U* Gate

- We have already proved that $CNOT = CX$



- Now, we want to calculate the matrix associated to $(C \uparrow)CNOT$



Swapping Roles in *controlled-U* Gate

- Since

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \begin{matrix} U_{00} = U_{11} = 0 \\ U_{01} = U_{10} = 0 \end{matrix} \quad \longrightarrow \quad (C \uparrow)CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

- $(C \uparrow)CNOT$ is Hermitian, i.e. $(C \uparrow)CNOT = (C \uparrow)CNOT^\dagger$
- Furthermore, $(C \uparrow)CNOT$ is unitary, i.e.

$$[(C \uparrow)CNOT][(C \uparrow)CNOT]^\dagger = [(C \uparrow)CNOT]^\dagger[(C \uparrow)CNOT] = I$$

Swapping Roles in *controlled-U* Gate

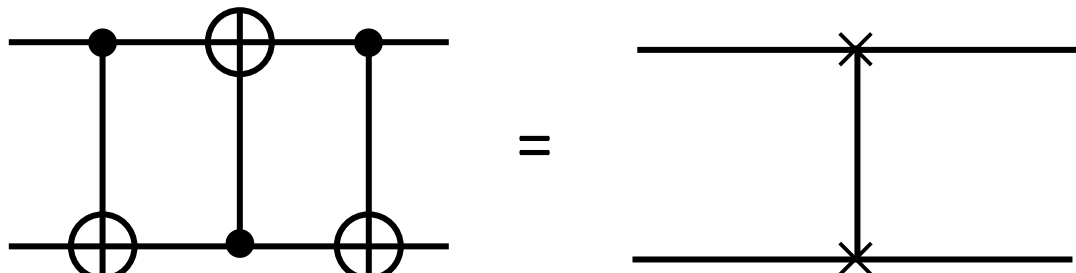
Proof

$$[(C \uparrow)CNOT][(C \uparrow)CNOT]^\dagger = [(C \uparrow)CNOT]^\dagger [(C \uparrow)CNOT]$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = I$$

Swapping Roles in *controlled-U* Gate

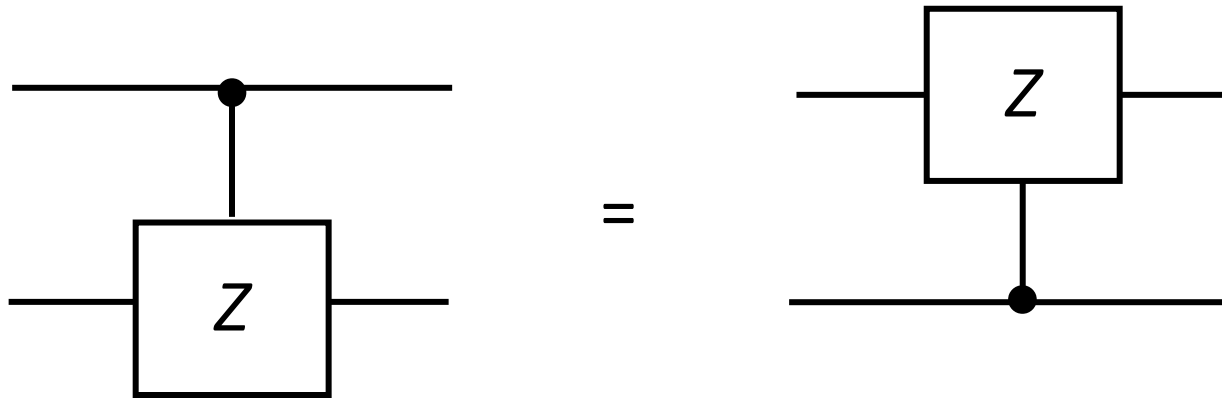
- It is easy to show that



$$CNOT \cdot (C \uparrow) CNOT \cdot CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = SWAP$$

Swapping Roles in *controlled-U* Gate

- Furthermore



- In fact, from $U = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ $\begin{matrix} U_{00} = 1 \\ U_{11} = -1 \\ U_{01} = U_{10} = 0 \end{matrix}$ $\Rightarrow$ $(C \uparrow)Z = CZ = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$